Control Voltage Converter

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This User's Guide is for the First, Second and Third Generation CVCs.
Screen images are from the 10.09 version of the Haken Editor
This Guide is also appropriate for 9.x versions of Continuum firmware and later.

Version 2.0

About this document

The current version of this document can be found online in the Support area of www.HakenAudio.com. We suggest new Continuum Fingerboard, ContinuuMini and Osmose owners read this guide concentrating on the interface for your specific instrument (Haken Audio Slim Continuum, ContinuuMini, EaganMatrix Module and/or Expressive E Osmose). Section 8 can be skipped if you have no need or interest to program the CVC. The knowledge you gain will save you time in the future and significantly enhance your experience with your instrument and CVC. After you finish reading this guide, please keep it available as a reference.

Note: This guide contains much more CVC programming information than the previous version and that may be useful to read, even if you have the older model CVC as they both function identically from the CV control and output perspective. This manual is considered 2.x as a previous manual exists for the older generation CVC which can be used with all the instruments discussed here.

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1. Overview of the Control Voltage Converter

The Control Voltage Converter (hereafter referred to as the "CVC") allows for the note on/note off (W Gate) polyphonic X (pitch), Y (front-back motion), and Z (pressure) outputs of the Continuum Fingerboard to be converted into control voltages (-10V to +10V max range – configurable to different voltage ranges through the Haken Editor). It also can be used in a similar manner connected to the Expressive E Osmose MPE keyboard, however, there Y represents an Aftertouch pressure component initiated after Z reaches its maximum level.

Note: This manual primarily discusses the current "low-profile" version of the CVC shown below, but the previous version of the CVC shown in Fig. 3 operates identically except for lack of TRS I^{2}C input.

Figure 1 - Current Low-Profile CVC – Dimensions: 38.1 x 15.3 x 4.4 cm (15 x 6 x 1.73 inches)

Figure 2- CVC with Optional Rabbit Ears for 19" Rack Mount

Figure 3- Previous CVC model - Identical in function (deeper and heavier with internal power supply)

The Continuum Fingerboard controls the CVC using the I^{2}C serial standard. The communication rate is 400 kHz and the interface is proprietary though available on request from Haken Audio (please raise a support ticket) if you want to try and integrate the CVC into a different host. To achieve this, the data output of the Continuum, ContinuuMini, EaganMatrix Module and Osmose is connected directly to the CVC's data input, avoiding Midi altogether, although in some cases the Midi port is used for the connection carrying the CVC's

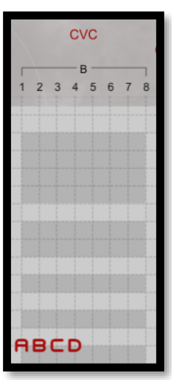
I²C data on unused pins. The CVC is specifically made for the Haken Audio instruments and the Osmose and will not work with any other device. A total of 16 unique control voltage (CV) streams/ports, typically divided into four logical channels/(port groups) of W, X, Y, Z are available from a single CVC.

When you connect your CVC to your Continuum Fingerboard, ContinuuMini, EaganMatrix Module or Osmose keyboard, the internal computer automatically identifies which CVC serial number is connected and intelligently generates customized I²C messages for your particular CVC that take into account your CVC's factory calibration data.

The CVC can be used in two ways:

(1) As a fixed configuration CV Interface using one of the standard CV voltage output definitions selected in the Haken Editor. (See Section 7 of this guide for operation information.)

(2) As a programmed CV interface using either Bank B or C in the EaganMatrix. Each user-defined CV output can be programmed based on finger W, X, Y, Z information, and can include processing by filters, delays, shape generators, and other EaganMatrix components. (See section 8 of this guide.)



Note: When playing the Continuum, ContinuuMini or Osmose from the fingerboard/keyboard and also playing through an External Midi interface, Midi data will be merged with the finger data to create CV output according the MPE Priority set. The same applied to playing the EaganMatrix module over CV while also sending it Midi note data. For the most enjoyable experience we recommend you use either one or the other input method at a time to avoid confusion.

1.1. Front Panel

Figure 4 - Front Panel of the CVC

The sixteen control voltage outputs are grouped into four sets of four CV banks/channels, each representing output of up to four simultaneous MPE channels (output according to current MPE priority set). When Standard CV Definitions are used with the CVC, each set of four outputs corresponds to one finger pressing on the surface; each finger has an associated Gate (W), X (pitch), Y (front-back – or Osmose Aftertouch), and Z (pressure). The red Gate (W) LED indicates when a finger is down, and the gate is active. A variety of Standard CV Definitions are available in the editor that are maintained in preset configuration, this is a default configuration:

- W Gate: 0v inactive, 10v active
- X: 1v/octave, with middle C at 0v Y: 0v to 10v (with 5v in the middle) Z: 0v to 10v

Depending on your analog synthesizer, you will have particular preferences for voltage ranges of each output on your CVC. Your Continuum will allow you to switch between Standard CV Definitions at any time through the Haken Editor and these definitions are stored with your preset when saved. Section 7 of this document details the Standard CV Definitions available.

Take care to avoid shorting CVC outputs to ground for extended periods of time, as this could cause the CVC’s output drivers to overheat. If you have trouble with this, or if you would like optional output protection resistors installed in your CVC, please contact Haken Audio.

1.2. Back Panel

Figure 5 –CVC back panel view

The current CVC model is powered by an external AC power supply supporting a voltage range of 100V-240V so this can be used in most all countries with the appropriate AC plug adapter (US and other country power plug adapters are supplied). Spec for the AC adapter:

- Input: 100-240VDC, 50-60Hz, 1.5A
- Output: 15VDC, 400mA

CVC I^{2}C Connection Options:

The current CVC has two I^{2}C connection options

- **Midi DIN connection**. In this case you must use a fully populated Midi Cable as the I2C uses unused pins. This connection is used for ½ and Full-size continuums and the Osmose.

- **TRS connection**. A standard TRS cable can be used, though one is supplied with the CVC. This connection is sued for Slim Continuums, ContinuuMini and EaganMatrix Module.

Note: Do not use both I2C connections at the same time connecting to two different Continuum or Osmose instruments. This can cause unexpected operation and possibly send the Continuum into a blue/purple blinking fault case.

<u>MIDI Din Thru Connection</u>

This is used if you connected to the CVC using a Midi Din cable and want to connect additional Midi devices to the Continuum or Osmose and play them over standard Midi (as the Midi data sent to the CVC will be passed through without the i2C signal in this case). It is not used if you connect to the CVC using the i2C TRS input.

Note: The two pins that are not used by standard Midi, pins 1 and 3, carry the i2c signals that are used to communicate between the Continuum and the CVC. The CVC does not use the standard Midi signals at all (pins 2, 4, and 5) except to pass them on passively to the Thru connector.

If you have a Continuum EaganMatrix Expander (CEE), connect your CVC to the CEE’s Data 4 connector, and connect all your Midi equipment directly to the Continuum’s Midi In and Out connectors. The CVC’s Thru connector will not be operational if you have a CEE; Midi Out is supplied directly by the Continuum Fingerboard’s Midi Out.

<u>Owners of the previous CVC</u>:

This unit does not have the TRS connection as it was intended for Continuums that only included the Midi DIN connectors. Thus, it cannot directly connect to the Continuums that require TRS input from the CVC. A MIDI DIN-to-TRS adapter must be created to connect to the previous model CVC. See Section 3 on ContinuuMini Connections for the cable/adapter pinouts (this information is also in the Continuum User’s Guide).

1.3. Cables

The CVC uses 1/4” TS (mono) jacks. For most Applications like Eurorack you will need to use 1/4” to 3.5mm Mono cables or cable adapters. Other modular form factors may require 1/4”-1/4” or even 1/4" to banana plug cables.

1.4. Mount

The CVC can sit on a table as it comes with rubber feet. You can also use the included rabbit ears to mount in a standard 19” rackmount enclosure (where you may have to remove the rubber feet).

2. Connecting the CVC to the Continuum Fingerboard

2.1. Slim Continuum Connection

Standard Connection: The CVC's I^2C TRS input port must be directly connected to your Slim Continuum Fingerboard's I^2C TRS output port using the provided mini TRS-to-TRS cable or a standard mini TRS-to-TRS cable should work as well. You will normally connect the Slim Continuum to your computer through the USB port (though the MIDI ports will work as well). Note the Slim Continuum has TRS midi ports that can be connected to other Midi devices using TRS-Midi DIN adapters provided with the Continuum. Thus, for the Slim Continuum, you will not have much need to use the CVC Midi THRU-port.

Figure 6 - Slim Continuum CVC Connection Diagram

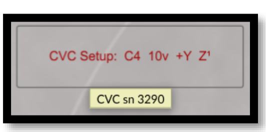
For Older CVC:

If you have an older CVC without the TRS CVC connection, you will have to make your own TRS-MIDI DIN CVC cable or a similar adapter to connect to a standard Midi cable. See the ContinuuMini section (section 3) below that has an example of the connection and the required pinouts.

2.2. Validating the CVC Connection

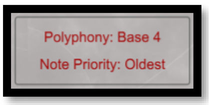
The CVC connection should be made automatically if the CVC and Slim Continuum are powered on and correctly connected. The blue light on the front panel indicates the CVC is powered, not connected.

An easy way to test if the CVC is connected is to go to Control Panel 3 in Haken Editor, hover your mouse over the CVC Setup control. If the CVC is connected as expected to the Continuum device or Osmose you will get a "CVC sn xxxx" connected message. You can connect the CVC while the Continuum is in operation, however, if your CVC is connected and not registering, try re-powering the CVC and/or Continuum.



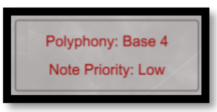
If the CVC is not detected, a "No CVC Connected" Message will appear. Note that if you remove the CVC connection, the CVC connected message may remain even though the CVC connection was removed. You can switch to another Standard CVC voltage option and back and you will then receive the not connected message.

Next, assuming you want to use the max CVC polyphony, in the editor set a preset for Polyphony 4 (not 4+) (see the Continuum User Guide for instructions for configuring polyphony). Your preset should have a setting as follows:



This will make sure that each time you press a finger one of the four banks of CV outputs trigger the red gate lights to your given polyphony and MPE priority. If you are set for more polyphony, fingerboard presses channels past 4 may not light up or may be interspersed with the 4 expected channels and you could be confused. Note that if you press two or three notes simultaneously, the order of the red lights may not be from left to right as Oldest round robin priority is set as the default.

Set your priority to Lowest. Now when you press polyphonically in sequence, the W and associated channel lights will light from left to right – lowest to highest channels played. In this manner you can predict what notes you are playing map to which CVC channels are output.



The next step is to configure the CVC to output the desired voltages to support the devices you are connecting to and understand CVC operation (see Chapter 7 for operating information).

The following four sections are specific to connecting the CVC to your Haken Audio or Expressive E instrument.

2.3. Half-Size and Full-Size Continuums

If you have non-Slim half-size or full-size Continuum, they connect to the current CVC using a fully populated Midi DIN cable from the Midi Out port of the Continuum to the I^{2}C DIN of the CVC, with optional use of CVC THRU port as needed if sending Midi to other Instruments or if using the Haken Editor with the Roland UM-One Midi adapter (required for Continuum firmware upgrades).

Figure 7- Half-Size / Full-Size CVC Connections – with and without Haken Editor

DSP Upgrade Note: For the proposed Half-Size or Full-Size DSP hardware upgrade (not currently available), a TRS Midi connection is planned that will require the Custom DIN-TRS cable or adapter (see section 3).

3. Connecting the CVC to the ContinuuMini

The ContinuuMini connects to the CVC for support of 1 or 2 voice output (as the ContinuuMini's surface only supports duo-tactic output).

To connect to the ContinuuMini:

- Use a TRS-TRS cable connected to the Pedal input of the ContinuuMini
- In the ContinuuMini Menu System set the Pedal I2C option to 1 (which means use the pedal input as I^2C CVC connection not a standard pedal). Remember to set this back to 0 if you wish to discontinue CVC use and want to support the standard pedal connection.
- Set your MPE Priority to Low or set Polyphony to 1 (for monophonic output) or Polyphony 2 (for duo-tactic output). Setting Lowest Priority will guarantee that if your Polyphony is greater than 2, the lowest 2 voices will only be output. However, if you also plan on playing through MIDI or plan on using your ContinuuMini as a MPE-to-CV converter you can set your Polyphony to 4 and your fingerboard data will be merged with incoming Midi data and sent to the available CVC ports.

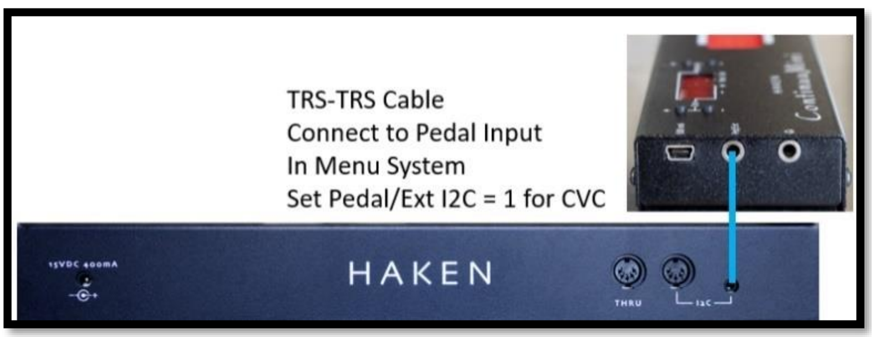


Figure 8 - ContinuuMini CVC Connection Diagram

All other CVC connection validation and configuration procedures are the same as for the Slim Continuum with the exception, as noted above, that you will only get a maximum of 2 channel CVC output if playing from the ContinuuMini surface.

Note that Midi input will be merged with surface output (or be output if no surface playing is present) so the ContinuuMini could support all four CVC channel output in this case and Polyphony should be set to 4 in this case.

<u>For Older CVC</u>:

If you have the older version of the CVC, you must use a custom TRS-DIN Adapter to connect to the CVC Midi port from the ContinuuMini Pedal port (set I^2C=1). Options for this and the required pinouts are detailed below.

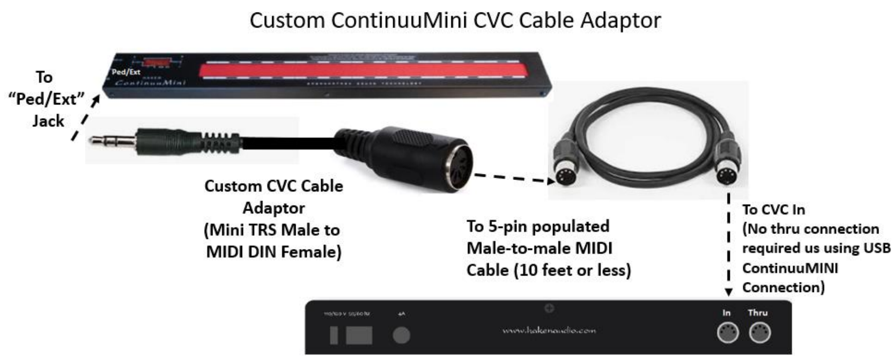
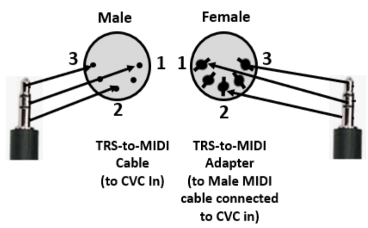


Figure 9- Custom CVC Connection for Older CVCs

Customer Adapter Pinouts are:



3.1. Using the uCVC by Evaton Technologies with ContinuuMini

Note: If you are only interested in playing CVC from the ContinuuMini fingerboard, a more cost-effective option is the uCVC by Evaton Technologies, especially if you plan on playing monophonic CV lines. Please contact Evaton Technologies in this case. The uCVC provides a single channel of CVC compatible W,X, Y and Z outputs. It has an additional F output (that outputs as long as your finger is contacting the ContinuuMini or Continuum surface). It also can daisy chain an additional uCVC to support duo-tactic mode.

Here's an example using the uCVC to play an EaganMatrix Module while splitting X and Z with mults to play in parallel a Stereo Oscillator (Chainsaw module) that takes three 1V/Oct inputs and has stereo outputs that get sent to a dual VCA before being sent to a Eurorack mixer. This will be a common method of playing multiple Eurorack modules from a ContinuuMini using a single channel of CV from either uCVC or CVC.

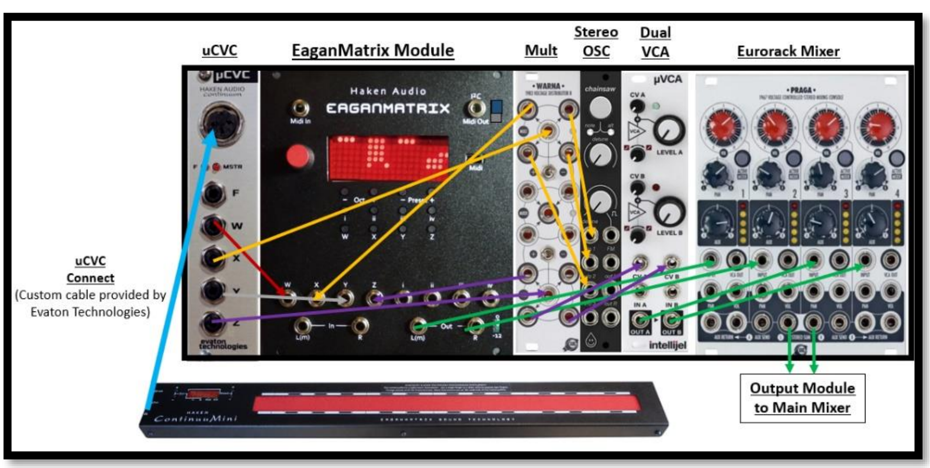


Figure 10 - Using uCVC instead of CVC for ContinuuMini/Monophonic Applications

4. Connecting the CVC to the EaganMatrix Module

The EaganMatrix Module is a bit of a different animal when it comes to CVC processing as it accepts CV inputs as well as outputting CV connected to a CVC. In this case the CV inputs (as well as any Midi inputs) will be translated and merged to the CVC outputs.

There are two CVC connection options:

<u>If Outputting CV from EaganMatrix Module with CVC</u>:

The EaganMatrix Module connects to the CVC in a similar fashion as the slim Continuum and ContinuuMini using a TRS-to-TRS cable. In this case:

- Connect the I^2C/Midi Out port of the module to the CVC TRS input port with a TRS-TRS cable.
- Set the module's I^2C dip switch in the up I^2C position. In this case the Midi output function of the module will be disabled. Remember to set the switch in the Down Midi Out position when you discontinue use of the CVC and wish to use this port for Midi output. The Midi Input port will be active regardless of the position of the I^2C/Midi Out Dip Switch.

Note: Though the module accepts 0 to 5V for Y and Z CV inputs, the CVC outputs retain their normal -10V to +10V support (depending on what CV output configuration you have selected or programmed).

<u>If playing the EaganMatrix Module from CVC Outputs</u>:

- Connect the desired W, X, Y, Z Channel from the CVC to the module's W, X, Y, Z CV inputs.
- If only the CVC is playing the EaganMatrix Module you may want to Set Preset Polyphony = 1 as the CV inputs are intended for mono-phonic output.
- If not using CVC programming on the device controlling the CVC, select CVC Setup Option #7 (W=5V, X=1V/Oct, Y=0..5V, Z=0..5V) in the editor for the controlling device.
- If using CVC programming on the instrument controlling the CVC, Gate can be >-1V, but limit Y and Z to 0..5V (if >5V is sent, module will limit to 5V). The same rules apply to sending any CV into the module for playing.

Configuration of the CVC is identical to that used for Slim Continuum and ContinuuMini, however the CVC in this case will be triggered by the W, X, Y and Z inputs to the EaganMatrix module which is typically a monophonic connection in addition to any MPE input.

<u>Use of CV Input with Midi</u>

When the EaganMatrix Module is connected to the CVC, you can output CV using the CVC from the module using either the module's CV inputs, Midi input (DIN or USB) or both CV input and Midi together and the information will be merged for CVC operation. MPE priority rules apply for CVC channel output.

When playing over CV, for example if you have Lowest Priority set with Polyphony 4 on the EaganMatrix module and you play a note through CV, Channel 1 of the CVC will output the data. If you hold that note and then play a note through the Midi port on the module, CVC channel 2 will output and if you play polyphonically through MPE CVC channels 3 and 4 will output if you play an additional three polyphonic Midi notes. Of course, if you set Priority to oldest (default) CVC output will follow the oldest round robin output sequence based on inputs received.

For example, you can connect a Linnstrument to the Module, set it to play Note per Channel on channels 2-5 (or less if desired), and with a CVC connected not only play the module polyphonically from the Linnstrument but also send the Midi-to-CVC polyphonic conversion of the Linnstrument output to other modules that accept CV inputs (see Use Case 3c in section 8 for details).

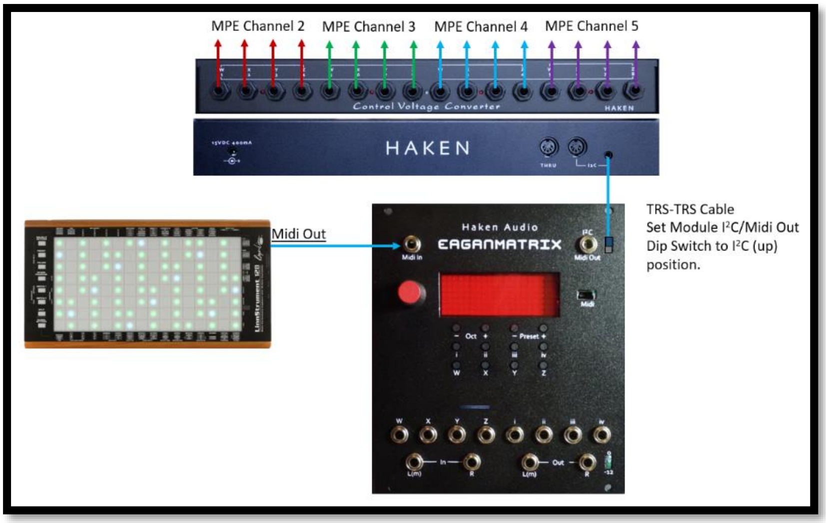


Figure 11 - EaganMatrix Module CVC Connection Diagram with Midi Input Source (here Linnstrument)

For Older CVC:

If you have an older CVC without the TRS CVC connection and want to connect it to the EaganMatrix Module, you will have to make your own TRS-MIDI DIN CVC cable or a similar adapter to connect to a standard Midi cable. See the ContinuuMini section below that has an example of the connection and the required pinouts.

4.1. Other CVC and EaganMatrix Module Use cases

4.1.1 Creating a Massive Multiple (Mult)

Playing the EaganMatrix Module from CV when connected to a CVC – Creating a Massive Mult

Of course, the need to use a CVC with EaganMatrix module is questionable if you want to play it from another Continuum Instrument or Osmose as you can connect directly to the module's Midi TRS Input port and play polyphonically over MPE. However, if you have an EaganMatrix Module, CVC and a CV input controller, the CV inputs will be routed to the CVC outputs and one possible use case of interest is if you set the Module's CVC setup as follows, the CVC can act like a 4-Channel W, X, Y, Z multiple.

- Set CVC Setup Option Layer Outputs (When Polyphony is 1 or 2)
- Set Polyphony 1
- Send CV into the module as normal. Note if your CV input device does not support Y (as is the case here) you can manipulate that directly on the front panel with the rotary (see Continuum User Guide for EaganMatrix module operation).

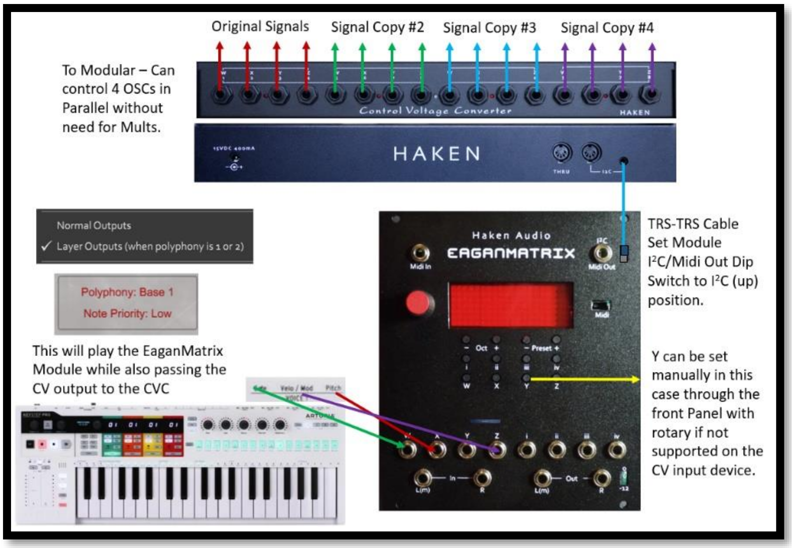


Figure 12 - EaganMatrix Module CVC Connection Diagram with CV Input Source

4.1.2 CVC with Multiple Source Connections – Midi Merging

Playing the EaganMatrix Module from CVC and Midi with Eurorack (Multiple sources)

This is another connection option if perhaps you have a Continuum, ContinuuMini, Osmose (or another MPE Midi Controller) and a CVC and you purchase an EaganMatrix module, perhaps to layer EaganMatrix audio output with that of your ContinuuMini or Osmose.

As noted, the EaganMatrix module can accept both CV and Midi input at the same time. The CV input is treated as if on channel 1 and the MPE input will normally be channels 2-9 if you are using the full 8 Voice polyphony. However, you cannot play more than 8 voices at a time.

In this case you will at minimum want to configure as follows:

- Set your EaganMatrix Module Polyphony to 8 (or 4+) if you want full voice support (note that this might cause some presets to issue reduce polyphony messages and set to another preset in that case if you need 8 voice support).
- Assuming the ContinuuMini is the instrument connected to the CVC, set the ContinuuMini preset you are playing to Polyphony 1 as you only want one voice to output to the CVC.
- Suggest setting Priority to Lowest on everything
- Set your Editor CVC Setup option for ContinuuMini to #7 that outputs 0..5V for both Y and Z.
- Connect CVC outputs to the EaganMatrix module for CVC Channel 1 W, X, Y, and Z outputs (CVC ports 1-4).
- When you play the Osmose (or other MPE instrument connected to the module’s Midi port the module will play polyphonic MPE data and that will be mixed with the ContinuuMini output when playing on ContinuuMini.

Note: If connected to the CV ports, suggest you do not play the module over Midi Channel 1 to avoid conflicts.

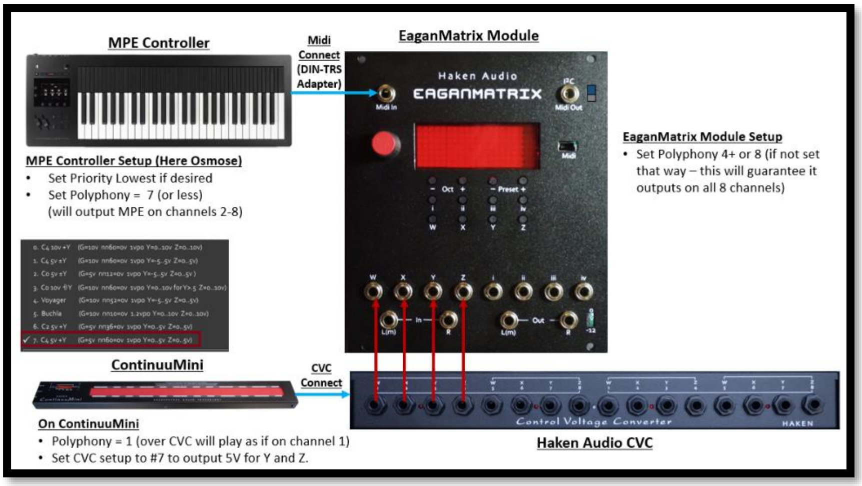


Figure 13 - Playing EaganMatrix Module from Osmose over Midi and ContinuuMini over CVC

5. Connecting the CVC to the Expressive E Osmose

The Osmose uses the CVC in a manner similar to the Slim Continuum as they both have the same DSP processing and polyphony, which is far more than the CVC can output.

To use the CVC with the Osmose:

- In the Osmose Global Midi I/O menu set Midi "din mode" to "5/5 cvc".
- Connect a fully populated Midi cable from the Osmose Midi Din Output port to the I^2C DIN input on the CVC.
- If Midi Thru connection is desired connect that to any Midi interface that you want to support downstream from the Osmose and connect the MIDI output from your downstream chain of devices back to the Midi Input of the Osmose. Note that the USB Midi for the Osmose can also be used for playing the Osmose or connecting to the Haken Editor and if so you may not require the Midi Thru connection. Remember to use the USB connection on the Osmose you will need a USB Midi host if no computer is connected.
- Suggest setting the Osmose preset polyphony to 4 if the goal is to play the CVC for modular control. You can play the internal Osmose presets to audio while outputting CV through the CVC and that may require more than 4 voices for audio output. You can set the Polyphony higher in this case the CVC will output the data according to the priority set but some channels will be dropped when playing polyphonically. So it's best to stick to 4 voice polyphony on the Osmose for CVC connections (this holds true for Continuum as well).

Figure 14 - - Expressive E Osmose CVC Connection Diagram

Although the current firmware version of the Osmose outputs MPE midi using Same priority, the CVC output will follow the Editor's MPE Priority setting if changed, however remember that Y will not output voltage until the Aftertouch pressure on Osmose is triggered – always after Z has reached it full value:

Once connected, the Osmose is configured for CVC operation identically to the Continuum devices.

<u>For Older CVC</u>:

Because the older CVC has a Midi DIN input port, connection to that will be identical to connecting to the current model. If you can find a used CVC this may be a cost-effective option as no special adapter is needed, just a fully populated midi cable.

Note: All older CVC models should be compatible.

6. Connecting the Previous Generation CVC

If you are an owner of the half-size or full-size continuum who already has the older generation CVC, this is perfectly fine for use with all new Continuum devices and the Osmose. See notes above for use with each device. Special adapters will be required in cases where a TRS CVC connection is required (see section 3).

When using a standard Midi cable to connect to the CVC, the CVC works best with a shielded Midi cable that has all 5 pins connected and has good connectors (such as high quality Neutrik connectors). Haken Audio has tested the Excellines Midi Cable, with a length of 10 feet or less. This cable is available in a variety of lengths from many on-line stores, including B & H.

6.1. xCEE Connection

If you are using an external Continuum EaganMatrix Expander, connect your CVC’s In port to the xCEE’s Data 4. If you use the CVC in tandem with Midi synths or Midi connections to a computer, connect your Continuum’s Midi Out/In to the Midi In/Out on your Midi synths or computer. If you have two CVCs, you can connect the second CVC to the xCEE’s Data 2 connector, for a total of 32 CV outputs.

Please contact Haken Audio support if you are having issues here.

7. Operating the CVC (Non-Programmed Use Cases)

The Continuum Fingerboard will generate output for the CVC based on finger movements on the playing surface. A single CVC can play up to four notes simultaneously. Normally you would configure your Continuum, ContinuuMini, EaganMatrix Module or Osmose to polyphony of 4 or less when you use the CVC. If you configure polyphony greater than 4, a single CVC will assume polyphony 4 and MPE Priority rotation will occur. Polyphony played grater than 4 will drop CVC output channels/voices based on current Priority set.

If you configure Polyphony 1 on your Continuum, the leftmost four Gate, X, Y, Z outputs on the CVC will be used. If you configure Polyphony 2, then the left two sets of four outputs will be used, etc. If you configure Polyphony 4 (or grater), all four sets will be used. Set your Note Priority to match your needs.

The order in which the new notes from the playing surface are assigned to Midi channels can be set in the Editor based on the MPE Priority you assign, defaulting to Oldest on Haken Audio instruments and Same on Expressive E Osmose.

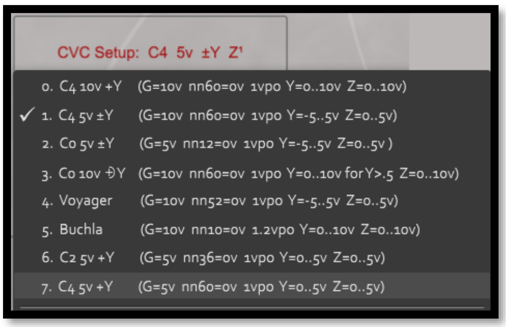
- **Oldest** – also called LRU (Least Recently Used) - assigns the new note to the Least Recently Used voice. This is the default for all Continuum-based instrument presets.
- **Lowest** - LCN (assign to the Lowest Channel Number). Here the CVC voices will be output in the order you press and release from Leftmost (lowest) channel to rightmost (highest). This mode is arguably the most useful as you can easily predict what CVC channel will be outputting based on the finger order you press. With oldest priority you will soon lose track of which finger is assigned to what channel especially if you play polyphonically. Also, when the CVC is controlling a monophonic analog synthesizer, it can be useful to use LCN, so that the first finger to touch the surface always outputs on the leftmost set of four Gate, X, Y and Z outputs on the CVC, and the second finger to touch the surface (after the first is already touching) outputs on the second set of four outputs, etc. In this way, the second finger can be used to consistently control some completely different parameters than the first. Or if you want to only output one set of CVC voices for monophonic support, set Polyphony to 1 (not 1+) and then no matter what the priority only CVC channel 1 will be output.

  For playing MPE plugins and instruments that are playing back audio using the same sound, it should not matter what MPE mode is set as Polyphony should be automatically handled correctly no matter what the Priority.
- **Same Pitch** – This is the default priority for Osmose MPE output. This keeps track of notes played up to the current polyphony and always outputs the same note on the same channel using oldest round robin scheduling. Because it repeats channels on the same note it is more predictable than Oldest for CV output mapping to fingers but you will eventually lose track of what channel is assigned to what finger as soon as you play more different notes than the CVC polyphony allows. In this sense it is really similar to Oldest priority.
- **High (High 1)** – Use only the highest Midi Channel within the current polyphony. For Polyphony 4 or greater this will always output on channel 4 regardless of what polyphony you play so you have to be careful as it might seem like some notes would be dropped. But it is useful for monophonic output when you will be guaranteed the highest channel will be output.
- **High 2 Channels**. This is like the previous option only the highest 2 channels will be output. It is useful when you play two voice polyphony and want to predict CVC output will always go to channels 3 and 4.
- **High 3 Channels**. This is similar to High 2 only the highest 3 channels in your polyphony will be output.
- **High 4 Channels**. This is similar to High 3 only the highest 4 channels in your polyphony will be output.

Note: Continuum Fingerboard's Mono and Split capabilities may be useful if you use the CVC together with standard Midi synths. If you have not done so already, please read about the Mono mode and Split configurations in the Continuum User's Guide. Basically, you can split the fingerboard or Osmose keyboard so that you output Midi below the split and Internal sound above (or vice versa). CVC output in this case will only be active for the Midi split. This allows you to play Internal sounds, possibly an accompaniment, below the split and play a melody over the CVC on a modular above the split.

7.1. CVC Standard Voltage Ranges

The Editor allows selection of the following CVC voltage configuration output options available in Control Panel 3 CVC Setup control. Different modular instrument brands accept different voltage levels and this will allow you to tailor the CVC voltage output for a desired instrument. You can also program your own voltage output using the EaganMatrix CVC programming option (see section 8).



Available Voltage options:

CVC Voltage Ranges		Gate Voltage	X Voltage	Y Voltage	Z Voltage
0	10v C4 +Y	10	1 per octave, 0 at nn 60	0 to +10	0 to +10
1	5v C4 +/- Y	5	1 per octave, 0 at nn 60	-5 to +5	0 to +5
2	5v C0 +/- Y	5	1 per octave, 0 at nn 12	-5 to +5	0 to +5
3	10V C0 Modified Y	10	1 per octave, 0 at nn 60	0 from front to middle, 0 to 10 from middle to back	0 to +10
4	Voyager	10	1 per octave, 0 at nn 52	-5 to +5	0 to +5
5	Buchla	10	1.2 per octave, 0 at nn 10	0 to +10	0 to +10
6	5v C2 + Y	5	1 per octave, 0 at nn 36	-5 to +5	0 to +5
7	5V C4 + Y	5	1 per octave, 0 at nn 60	0 to +5	0 to +5
	Each of the above is available with Linear Z or Squared Z slope in CVC Z outputs (4,8,12 and 16).				

7.2. Z Scaling – CVC Setup Options

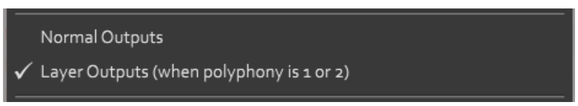
The Editor allows selection of two different Z pressure scalings for CVC outputs based on pressure detected from the input device (defaulting to Linear). This is independent from the internal Z pressure programming in the matrix applying to internal presets.

- Use Z output as a linear function of pressure
- Use Z output as a squared function of pressure.

Note on Osmose has its own internal sensitivity pressure control scaling options that may conflict with Editor settings as they can be set independently. You may have to do some experimentation here to determine what settings work best for your application.

7.3. CVC Layering – CVC Setup Options

The CVC can output the same control signals on multiple outputs for doubling outputs. If Layering is selected in the Continuum Editor’s CVC menu the following will apply:

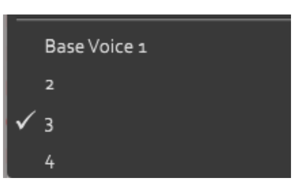


If Polyphony is 1 and Layering is selected, the Gate, X, Y, Z for each note will be available on all four sets of CVC outputs.

If polyphony is 2 and Layering is selected, the Gate, X, Y, Z for each note will be available on the first and third, or the second and fourth, sets of CVC outputs depending on which voice is playing.

7.4. CVC Base Voice – CVC Setup Options

The control signals generated by the CVC can be shifted to different outputs if Base Voice 2, 3, or 4 is selected in the Continuum Editor’s CVC menu. For example, if you have Polyphony set to 4 and select “Base Voice 2”, the CV outputs will occur on Channel 2 (ports 5-8), Channel 3 (ports 9-12) and Channel 4 (ports 13-16). Channel 1 (ports 1-4) will never be output. The similar limits will be applied of you set “Base Voice 3” – Channels 1 and 2 (ports 1-8) will never be output. Note Priority rules apply for channels that are output.



7.5. Controlling the CVC from a Keyboard or Sequencer

As noted, the CVC will output control signals originating from data fed to the Continuum Instrument or Osmose Midi in port.

By default, the CVC will play a merge of the Midi messages coming in on the Continuum's Midi In and notes resulting from fingers on the playing surface. Avoid having both the playing surface and the Midi In using the same Midi channel, as the merged Midi messages will invariably interfere with each other.

In this manner if you play the Continuum or Osmose from an external Midi sequencer, those inputs will be translated to CVC outputs if a CVC is connected.

7.6. Osmose Sequencer Output

The Osmose Sequencer will output CVC control voltage based on the sequencer settings but output may be a bit erratic. Use Polyphony = 1 for best results. You can send this into the EaganMatrix module if you have one and sequence it with Osmose over CV.

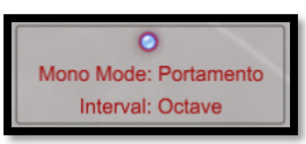
- W Gates should track Channel pressure motion
- X should map to Osmose key note. Bend will apply to sequenced notes if you apply it.
- Y will be output as expected
- Z will track the pulsing of the sequencer output but scaled with pressure so you can vary the volume of your sequence with Z as expected.

Note: If you want to run the Osmose sequencer through Midi on the Continuum, ContinuuMini or EaganMatrix Module, set the DIN mode back to its default "1/5default setting" to avoid Midi loops.

7.7. Mono Mode and Pressure Weighted Portamento Output

Set Polyphony – 1 for Mono mode operation. It will not work as expected for higher polyphony settings.

If you set Mono mode on Continuum-based instruments notes of the portamento will occur on the current channel you started it on to the interval set and X bend scaling will be output as expected. Osmose will also output Pressure Weighted Portamento (which is the same as Mono Mode Portamento in the editor) as expected for X.



On Continuum and ContinuuMini portamentos will be output on X to the CVC as expected so you will be able to send a full range portamento through X with bend being translated to the appropriate 1V/OC X scalings.

7.8. Bend and Rounding with CVC output

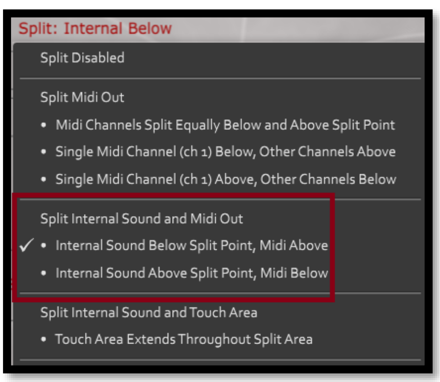
CVC X pitch output will follow bend (relative 1V/Oct ratio output) and rounding scaling on Haken Instruments. Note that the more you round the less effect bending notes will have. If you want the full effect of vibrato set rounding off or at a low value.

Note: Osmose does not currently support Rounding as expected when set in the editor. We suggest you always leave it off in the editor. Osmose will round pitch on the key. That is when full rounding is set, Osmose will raise the pitch 1/2 step when you bend the key all the way to the right and lower it a step when you bend the key all the way to the left. If you want rounding on the Osmose you should set bend appropriately in the Osmose menus.

7.9. MIDI Split Mode and CVC Output

When you set a Midi Split in the Haken Editor allowing the internal sound to be output on one side of the split and Midi on the other, the CVC output will follow the Midi split.

The Split Internal Sound and Midi Out option is the more useful case when CVC is connected.



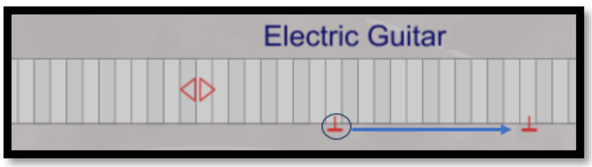
The Split point is set in the editor by moving the two facing triangles to the desired split position.

Note: This will work for Osmose but you cannot set a split on the Osmose itself as of firmware 1.06. You need to use the Haken editor to set the split. This may change in future firmware versions.

7.10. Transposing CVC Output

The CVC's X/pitch 1C/Octave scaling output will follow the current Octave/Transposition setting. This function must be set in the Haken Editor or use the Octave buttons on instruments that have them for octave transposition. See the Continuum Manual for more information on transposing.

The inverted T on the Haken Editor's Surface Display represents where Middle C lies on the surface (or associated key/note on Osmose). So as in the example below move Middle C up an octave, any note you play will in effect be transposed down an octave. The CVC output will follow this transposition. You can transpose to any interval this way.



Note: Osmose currently as of firmware 1.06 does not allow transpositions to be set in the menu system. You must use the Haken editor as described above. This may change in future firmware versions.

7.11. Multi-tracking the CVC from a Sequencer

It is possible to control the CVC via multiple Midi performances. In doing these types of Midi overdubs, it is quite conceivable to overload the Midi capabilities of the external DAW sequencer, as the DAW may have problems merging multiple Continuum originated Midi streams into a single Midi output stream. Therefore it is usually advisable to do overdubbing in a more traditional way, by recording the output of the analog synthesizer as an audio track, and overdubbing with additional audio recordings into the DAW.

However, to do Midi overdubbing, below is the sequence of events to record and playback two monophonic CVC tracks:

(1) Configure the Continuum with "Polyphony 1" and Note Priority to "High 1". Also disable Midi routing from Midi In to Midi Out.

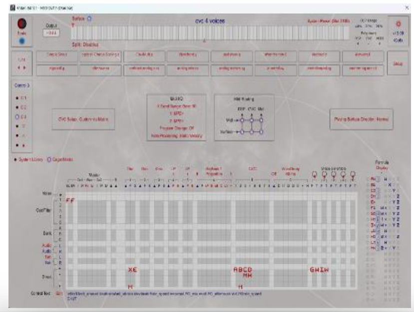
(2) Play the first monophonic track; record it into your Midi sequencer as you play in on the CVC. This first track will be on Midi channel 1, and use the leftmost four outputs on the CVC.

(3) After you have finished playing the first track, configure the Continuum to Polyphony 2.

(4) Record the second track, as you play back the first track from your sequencer, and record the new notes into your sequencer. The new notes will be on Midi channel 2, and will come out on the second set of four outputs of your CVC. The previous track is on channel 1, and will come out on the leftmost set of outputs on your CVC.

(5) Now you can play back the two recorded tracks together through your CVC; but don't touch the playing surface as you do so, because new notes will interfere with those two tracks (unless you set the Continuum's polyphony to higher than 2).

8. Preprocessed Control via the EaganMatrix



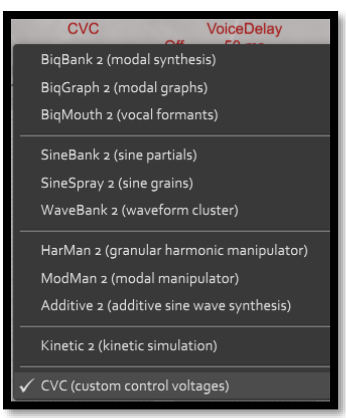
In controlling an analog synthesizer, it is often musically useful to able to have multiple source influence over the shape of a single voltage control that is being sent to a single synth CV input. For instance, a simple common two source synthesizer mapping would have (1) low pass filter cutoff point controlled via pitch (filter opens as pitch increases) and (2) volume (the filter opens as the loudness increases).

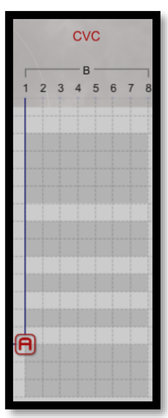
Using the Continuum Fingerboard's built-in synthesizer, the EaganMatrix, the CVC can have extensive and complex blends of unique finger controls mapped to its control voltage outputs. When "CVC" is selected in Bank B or Bank C, the Standard CV Definitions (Section 7.1) are not used; instead, the EaganMatrix allows you to create up to eight unique formula combinations. User-defined formulas directly specify the voltage value for up to 8 CVC outputs. The CVC will have 2 voice polyphony, with 8 user-defined control voltage outputs per voice. The CVC will have 4 voice polyphony, if you define only 4 control voltage outputs per voice and leave the last four columns blank. For details, please read the EaganMatrix User's Guide. Also check out the CVC category in the Editor's Preset selection menu for some simple programmed examples. The system CVC category (on Continuum) contains starting-point EaganMatrix presets for controlling CVCs. More complex examples (one analyzed here) can be found in the Osmose Utility category.

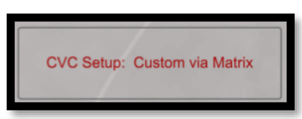
The basics of CVC programming will be presented below followed by a number of programming use cases:

8.1. Configuring Basic CVC Programming - Setup

If you select CVC option from either Bank B or Bank C, the CVC will become available for programmed output:



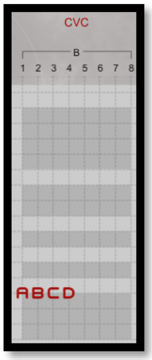


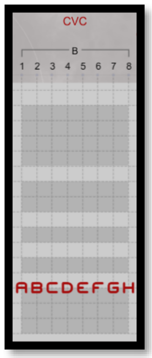


Once selected, the CVC Setup status will change to "Custom via Matrix" (you may in some cases need to enter a formula to see the change in status). At this point the standard CVC output configurations will be ignored and you can then program your own W, X, Y and Z formulas to output custom voltage configurations and/or complex programmed CVC outputs, including SG-based patterns.

8.2. Understanding CVC Port Mapping When Programmed in Matrix

You can specify one or two sets of CVC port formula configurations in the Matrix:





Normally CVC outputs 1, 2, 3 and 4 are programmed to map to the expected functions as displayed on the ports 1=W, 2=X, 3=Y and 4=Z. The similar relationship applies for ports 5-8 where 5=W, 6=X, 7=Y and 8=Z. But note this does not have to be the case. You could program X pitch to come out the Z port, etc. but this is counter-productive and confusing and it is recommend that you program W, X, Y and Z as normally used/expected.

Note that you can only program either Bank B CVC or Bank C CVC, not both. For example, if you do not use a delay in Bank C and use both Banks A and B for other purposes, Bank C can be used for CVC programmed output. If you need a delay and want CVC programming you need to reserve Bank B for that purpose.

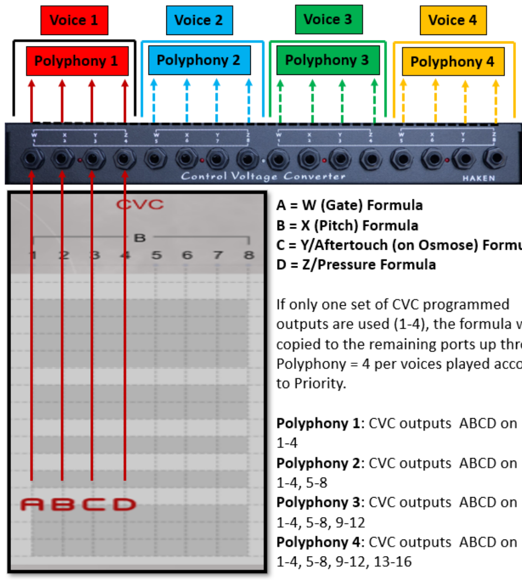
When programmed, CVC ports (CVC matrix columns 1-8) map to the 16 CVC output ports as follows. The Polyphony set in the preset affects the port mappings:

If groups of W, X, Y and Z are always used as units it can be easy to think of each group of 4 ports as a CVC channel with 4 possible channels being supported (Channel 1=Ports 1-4, Channel 2= Ports 5-8, Channel 3=Ports 9-12, Channel 4= Ports 13-16).

The CVC output port mappings are as follows:

Only Ports 1-4 are programmed in the CVC: (outputs copied per current Polyphony and Priority setting)

- **Polyphony 1 set**: ABDC (formulas above) output on ports 1-4
- **Polyphony 2 set**: ABCD formulas output on ports 1-4 and copied to ports 5-8
- **Polyphony 3 set**: ABCD formulas output on ports 1-4, copied to ports 5-8 and 9-12
- **Polyphony 4 set**: ABCD formulas output on ports 1-4, copied to ports 5-8, 9-12 and 13-16



A = W (Gate) Formula
B = X (Pitch) Formula
C = Y/Aftertouch (on Osmose) Formula
D = Z/Pressure Formula

If only one set of CVC programmed outputs are used (1-4), the formula will be copied to the remaining ports up through Polyphony = 4 per voices played according to Priority.

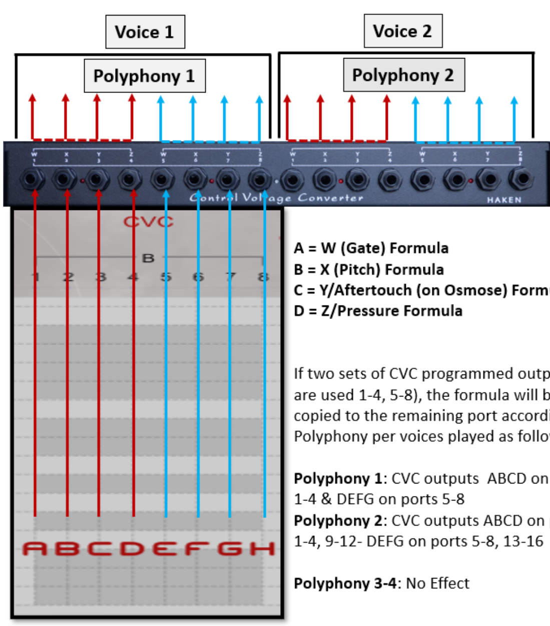
Polyphony 1: CVC outputs ABCD on ports 1-4
Polyphony 2: CVC outputs ABCD on ports 1-4, 5-8
Polyphony 3: CVC outputs ABCD on ports 1-4, 5-8, 9-12
Polyphony 4: CVC outputs ABCD on ports 1-4, 5-8, 9-12, 13-16

Only Ports 5-8 are programmed in the CVC – Not Allowed

You cannot program just ports 5-8. No CVC output will be created. If you program ports 5-8 you must also program ports 1-4.

All Ports 1-8 are programmed:

- **Polyphony 1 set**: ABDC output on ports 1-4, EFGH output on ports 5-8
- **Polyphony 2 set**: ABCD formulas output on ports 1-4, 9-12, EFGH output on ports 5-8, 13-16
- **Polyphony 3 set**: No additional effect
- **Polyphony 4 set**: No additional effect



A = W (Gate) Formula
B = X (Pitch) Formula
C = Y/Aftertouch (on Osmose) Formula
D = Z/Pressure Formula

If two sets of CVC programmed outputs are used 1-4, 5-8), the formula will be copied to the remaining port according to Polyphony per voices played as follows:

Polyphony 1: CVC outputs ABCD on ports 1-4 & DEFG on ports 5-8
Polyphony 2: CVC outputs ABCD on ports 1-4, 9-12- DEFG on ports 5-8, 13-16

Polyphony 3-4: No Effect

8.3. Effects of MPE Priority on CVC output in Relation to Polyphony

For example, if Polyphony is set to 4 and Priority is Oldest, if you play monophonically, notes will cycle between the four CVC channels (ports 1-4, 5-8, 9-12 and 13-16) according to the oldest round robin sequence.

For example, in this setting a 4-note chord is played with notes coming in staggered low to high. You can see the CV output gating in the Oldest Round Robin order in this scope output. You really can't tell which voice starts on which channel using Oldest priority.

Now priority is reset to Lowest and playing the same note sequence low to high, it's easy to determine which voice was played by what finger, lowest to highest channel. This could be valuable when sending multi-channel CVC output to different modules in a predictable manner from channels 1 thru channel 4. If you want your CVC channel output to rotate between channels use Oldest. If you want it to be repeatable per channel as you play your controller, use Lowest.

Note that if you are controlling the CVC from an Osmose, Continuum or ContinuuMini and connecting the CVC output to the EaganMatrix module's CV inputs, it's a good idea to set the Note Priority the same on both.

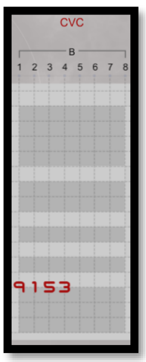
8.4. Programming Basic CVC Formulas

Important Note: This section assumes rudimentary EaganMatrix programming knowledge. See the EaganMatrix manual for more information. If you have no desire or need to program the CVC, you can skip the rest of this section.

Using Constants in CVC Programming

Constant values (as well as formula values programmed for W, X, Y and Z) should match the desired voltage outputs desired (in volts) in the range -10.0 to +10.0 (though the actual voltages output for the maximum range may be -10.01 to + 9.97 (or so). So don't assume a strict 10V output for gate as it could be a tiny bit off as noted. Also don't worry If you exceed the maximum or minimum range by mistake in your programming. The CVC has protection and will not exceed the maximum limits.

Consider the following CVC programming using constants:

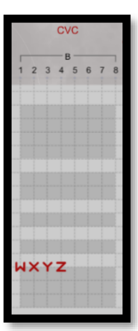


This will set:

- **W** (Gate) = 9 Volts
- **X** = 1 Volt (Assuming you are sending to a 1V/Octave OSC input, this would map accordingly to pitch – normally an octave about zero point). But you are not limited to outputting 1V/Octave scaling. You can use formulas to output a variety of different X scalings.
- **Y** = 5 Volts
- **Z** = 3 Volts

Using EaganMatrix Default Formulas in CVC Programming

Normally, formulas are used instead of constants and those formulas can change based on finger movement. You can use the fixed W, X, Y and Z formulas as the simplest programed example. The output will be different depending on if you are playing the CVC from Continuum/ContinuuMini or Osmose as noted below and you can see that in genera using the fixed formulas are not that useful:

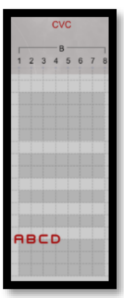


Understanding Output of the Fixed formulas if used in CVC columns.

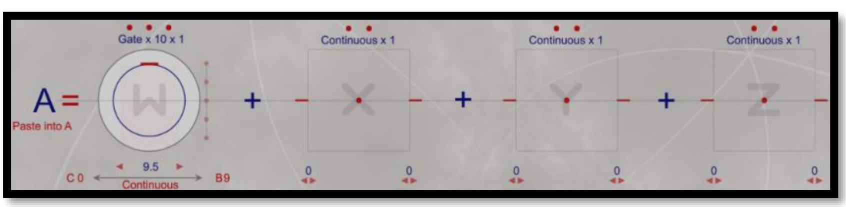
- **W** = 1 Volt that will be output as long as you are pressing the fingerboard or depressing an Osmose Key (might actually be just slightly less than 1V if you scope it). W is then not that useful unless you need a low gate voltage.
- **X** = Raw Voltage at key position scaled 1/V Octave but with zero Point at bottom of the fingerboard. In this sense X is not that useful in normal programming where you will want to scale by octave but have the zero point (normally mapping to middle C) be C4 or C5.
- **Y** = 0 to 1 Volt linear scaling. On Continuum or ContinuuMini Y retains its value at current front to back position at the point you lift your finger off the fingerboard. On Osmose Y will always return to 0V when you release the key.
- **Z** = 0 to 1 Volt squared scaling increasing as you press the fingerboard or Osmose key and returning to zero when you release your finger.

Obviously these values mapping to the standard W, X Y and Z constants are not optimal values for most modular applications. Instead set W, X, Y and Z with associated EaganMatrix formulas mapping to more desired values. Here you can set normal ranges with formulas and also if desired make use of all of the available W, X, Y and Z transfer functions to create custom voltage output not available in the seven fixed CVC voltage configurations discussed above:

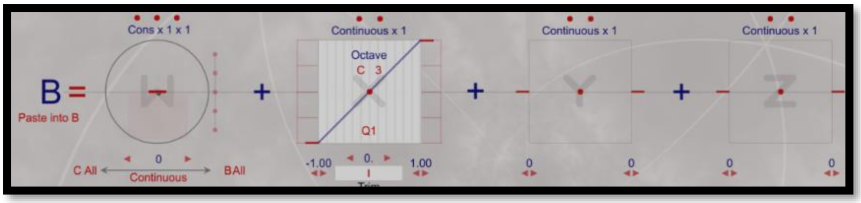
For example, the formulas A, B, C and D can be applied to the first four CV outputs per the Polyphony 1 configuration above to create the following custom voltage outputs (as one of an almost endless number of possibilities):



Gate On 9.5 V



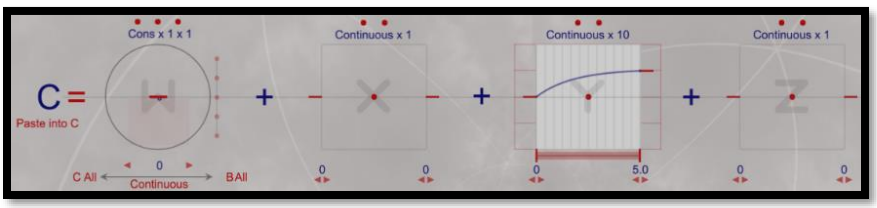
X = Pitch in correctly scaled 1V/Octave (Zero voltage point = C3, Quantized to the Semitone – Q1)



Zero point is normally mapping to Middle C on Continuum and the point around which you can scale X in the positive and negative directions. In terms of CVC it is the X scaling point where voltage output is 0. From there 1V/Octave scaling applies +/- around that point. Now of course if you are sending the X CV output to a modular oscillator you normally would fine tune that anyway to the desired pitch.

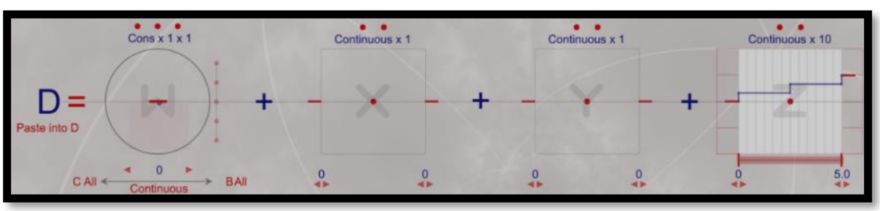
While the formula can also specify quantization to a number of intervals, the most common being a semi-tone, quantizing of that nature is inherent in Osmose output vs. on Continuum where voltage is continuous as you move your finger across the fingerboard so quantizing for Osmose would not be that useful in this case.

Y = 0 to 5V (using Square root transfer function)



With Y formulas you must specify the transfer functions desired. The fixed Y function uses linear scaling shelved a bit on each end. Remember Y on Haken Instruments maintains it last value unless it is set with a formula back to zero when finger is lifted off the fingerboard. On Osmose it returns to zero by default.

Z = 0 to 5V (but using a 3-step discrete transfer function)



With Z formulas you must specify the transfer functions desired. The fixed Z function uses a squared output.

This illustrates a fraction of the complex custom control voltage output formulas that can be created. In some sense, CVC programming allows you to create an arbitrary CV functional generator. Remember Z always returns to zero when a finger is lifted from the fingerboard/keyboard.

8.5. CVC Programming Use Case 1 – Continuum

This is a simple use case where all 8 ports are programmed with very simple formulas, providing a different use case than is available from one of the seven fixed CVC options. Here the CVC is programmed so that first set of four ports outputs 5V scaling and the second set of four ports output 10V scaling.

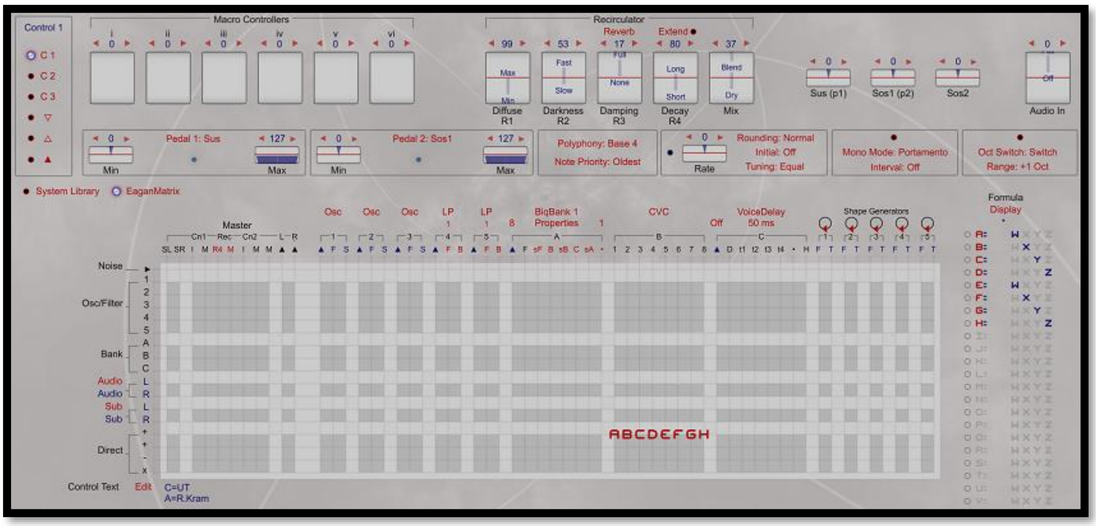


Figure 15- Continuum CVC Use Case using all 8 programming CVC columns

Programs ports 1-8. Outputs on CVC channels 1-4.

When 8 ports are programmed, CVC output will follow Note Priority with columns 1-4 for formulas A, B, C and D output on ports 1-4 and 9-12 and columns for formulas E, F, G and H output on ports 5-8 and 13-16.

This preset (or something like it) might be useful if you have some modules that need 5V scaling and others that need 10V which you can output in parallel with the same Continuum performance without need to rescale on your modular.

This preset does the following:

Ports 1-4 programmed as:

- W Gates CV at 5V
- X set at 1V/Octave, Zero point – C4 (Octave scaling)
- Y set to output 0-5V Linear
- Z set to output 0-5V Linear

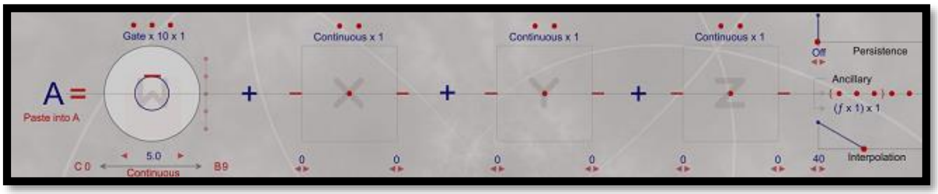
Ports 5-8 programmed as:

- W Gates CV at 10V
- X set at 1V/Octave, Zero point – C3 (Octave scaling) – One octave higher than port 2
- Y set to output 0-10V Linear
- Z set to output 0-10V Linear

CVC Channel 1 (Programmed CVC ports 1-4)

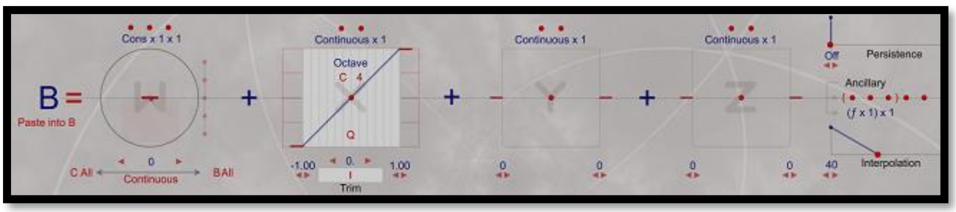
Z Gate CV Output Port 1:

On fingerboard gated press, 5V CV Gate will be output.



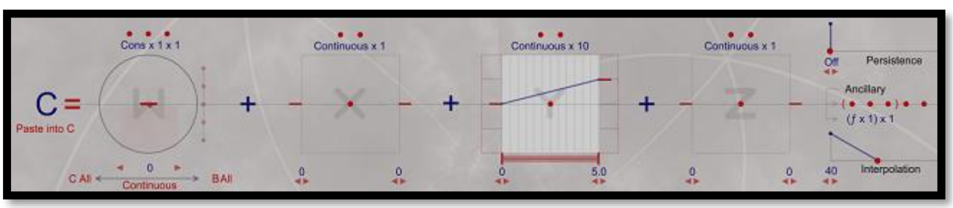
X Pitch 1/V Octave CV outputs on Port 2:

X CV will output 0V at C4. Since this is Octave (not KHz frequency) scaling C5 would output as 1V, C6 as 2V, C3 as -1V, etc. If sent to an OSC that needs frequency adjustment use a tuner to set the frequency that results when Continuum surface presses C4 to desired frequency (for example 261.63 Hz for equal temperament A-440 tuning) and then Continuum and your oscillator should be properly aligned for output frequency at 1V/Oct scaling mapping to the desired fingerboard position (or Osmose key if used instead).

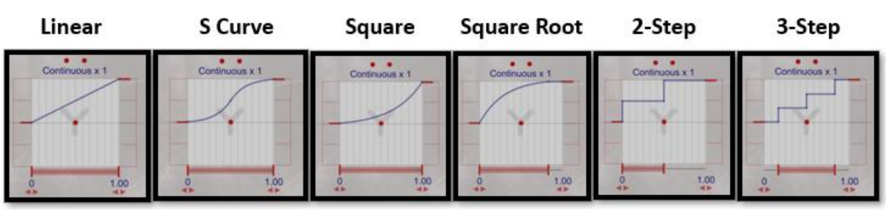


Y CV output on Port 3:

Simple Y output 0V-5V with linear transfer function and no shelving. You could set this to any of the Y shapes, including the stepped shapes and CV will follow the function.

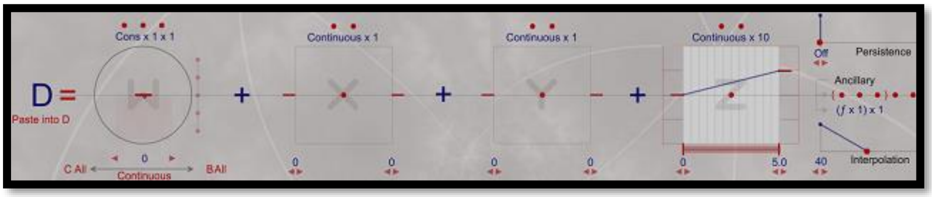


Y Transfer functions. In addition to the standard Linear, S, Squared Root and stepping options you can now also associate Y arbitrary voltage functions using graph offsets. (Note: X and Z use the same transfer function options).



Z CV output on Port 4:

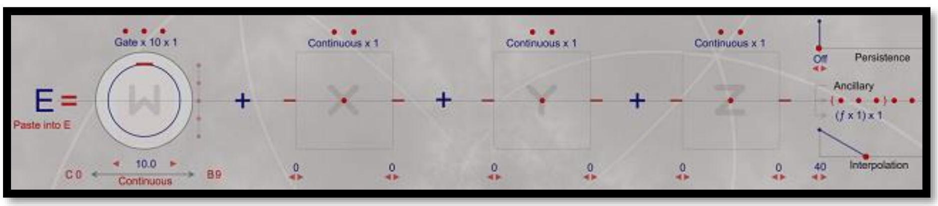
Simple Z output 0V-5V with linear transfer function and no shelving (default Z would normally be squared function for example). Of course, you could set this to any of the Z shapes, including the stepped shapes and CV will follow the function.



CVC Channel 2 (Programmed CVC ports 5-8)

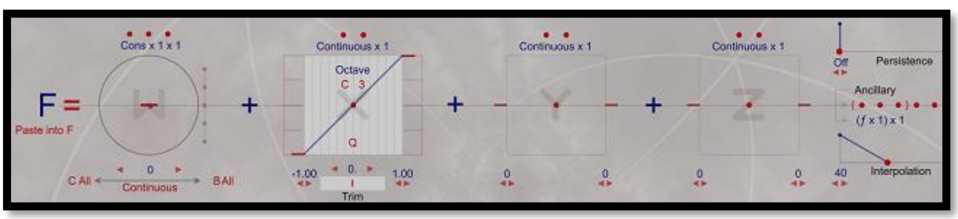
Z Gate CV Output Port 5:

The same as for port 1 only Gate output is now 10V.



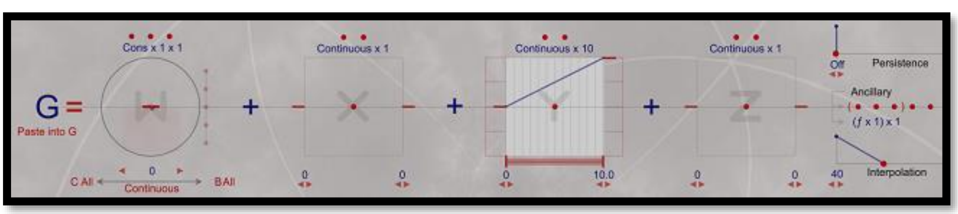
X Pitch 1/V Octave CV outputs on Port 6:

Similar pitch output as for port 2 only now zero point is set for C3, so in effect this is scaled up an octave as C3 will now output 0V and C4 that was 0V on port 2 will not be 1V (an octave higher) when played and connected to this port.



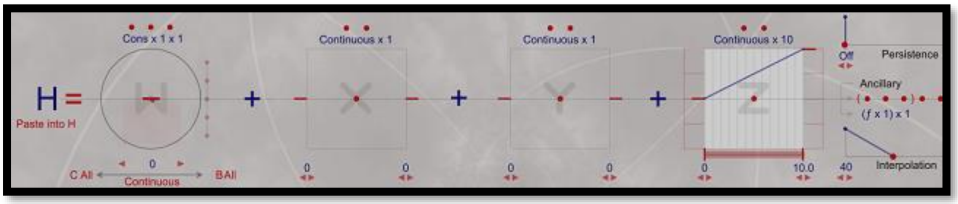
Y CV output on Port 7:

The same as for port 3 only Max Y output is now 10V.



Z CV output on Port 8:

The same as for port 4 only Max Y output is now 10V.



8.6. CVC Programming Use Case 2 – ContinuuMini

Programs ports 1-4. Outputs on CVC channels 1 and 2.

Since the ContinuuMini supports duo-tactic output set CVC preset polyphony=2 for best support or if you want to play monophonically set Polyphony 1. Priority is also set to Lowest so that normal monophonic playing will always output on the first CVC channel and the duophonic line will come in on the second channel assuming no polyphonic voice switching is performed.

In this case we'll demonstrate programming CVC in Bank C instead of bank B (you can do one or the other, not both)

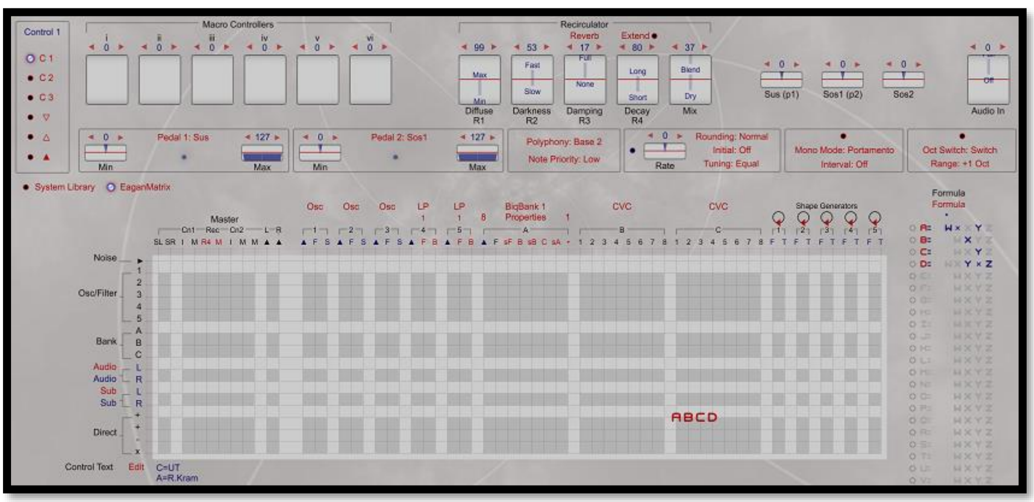


Figure 16- ContinuuMini CVC Use Case using 4 programming CVC columns

This preset does the following:

W will gate at 5V when ContinuuMini is pressed and Y is beneath Mid point. If Y is above midpoint and a note is pressed the resultant gate will be 10V.

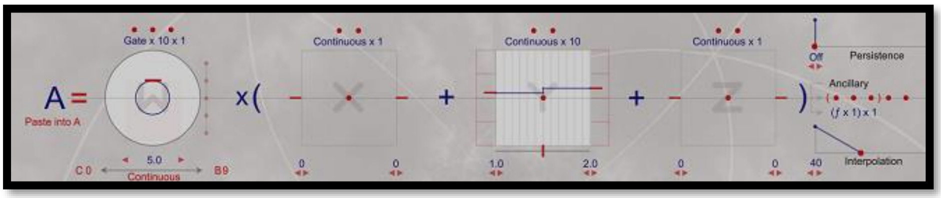
X will be scaled two octaves to one (two octaves worth of 1V/Octave output will be compressed and output when playing one octave on the fingerboard).

Y will be set as a two step function. 0V will be output on the lower half of Y and 5V on the upper half.

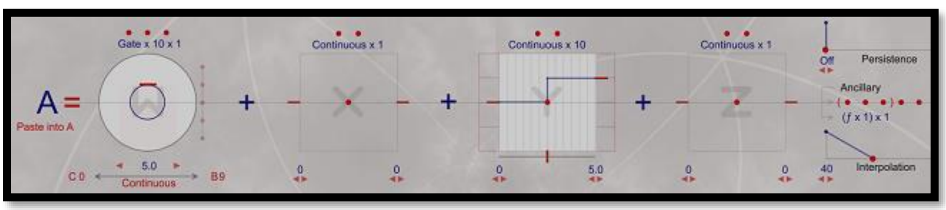
Z will be set as its normal squared function to output 0 to 5V while Y is at Min. As you increase Y an additional 5V will be added at top of Y so at max pressure and top of Y ca. 10V will be output

W Gate CVC output:

W is set to a gated constant of 5.0V. This is multiplied by a two-step Y function where bottom to Mid Y is 1.0 and Mid Y to top of Y is 2.0. Thus, when the fingerboard is pressed bottom-to-Mid of Y, gate will be output as 5V. When the fingerboard is pressed mid-to-top of Y gate will be output as 10V. Note in this case if you play and move across the Y midpoint in either direction you will change the gated value that is output until you lift your finger. So this is a special case output but it shows how flexible you can be when creating formulas that do not just have to include the value you are outputting (W in this case) in formula creation. You could have used W, X, Y and Z in formula A for a complex relationship. The output will still be from the CVC port associated with formula A (W in this case).

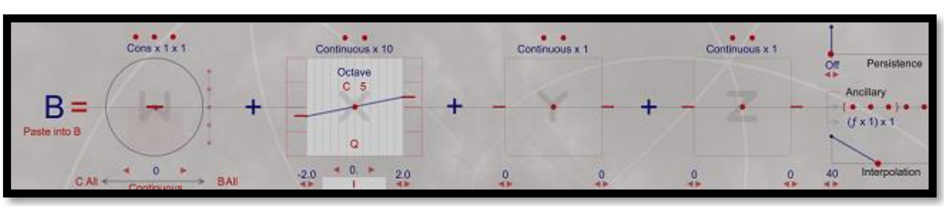


This uses a multiplication scaling to produce the desired value. In the example above, if you lift your finger, W goes back to zero because the Y value will be multiplied by the W Gate=0 value. If an addition of Y was used to try and create a similar output, a different voltage effect would be generated because Y retains its current value when you lift your finger. With the following formula if you press at bottom of Y, Gate will be output at 5V and if you press at top of Y gate will be output at 10V. But if you lift your finger at mid-to-top of Y, the formula value will not go back to zero as Y is still 5V. You will need to press again beneath the Y mid-point to set Z back to 0V. Not the effect we intended where lifting the finger is expected to set W back to 0V.

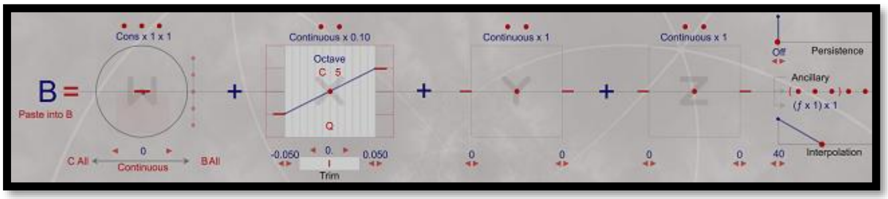


X CV outputting compressed octaves:

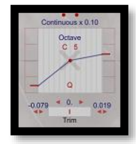
Formula B uses "Result Proportional to Octave" (Octave) mode to output X 1V/Octave scaling around the Zero point here set to C5. This maps fingerboard or keyboard notes as expected vs using the "Convert Result to kHz" option that would be used in many places to set an EaganMatrix frequency which is not desired for CV output in most cases as 1V/Oct scaling is expected in most cases. 0V will as normal be output at the Zero point, but now the scaling on X is set to a range of -2.0 to +2.0 instead of the normal default -1.0 to +1.0 range. In this case when playing C6, the X CV output will be 2V instead of the normal 1V (in 1V/Octave scaling) . At C7 it will be 4V. Thus you will play over a range of two octaves when you play only one on the ContinuuMini.



Note we could have done the opposite and expanded by an octave by setting X to a range of -0.5 to +0.5. Here Zero point at C5 will still be 0V but not C6 will be output as 0.5V and C7 as 1V thus requiring you to play a range of two octaves on the fingerboard to cover a range of one octave.



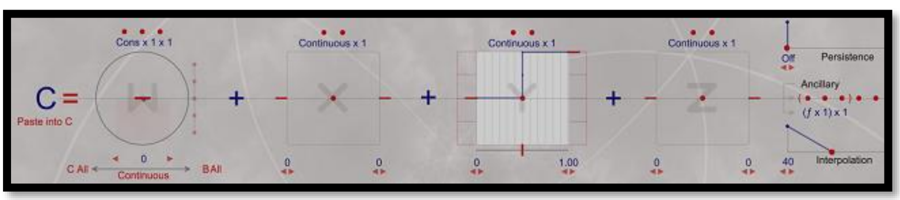
You are also not limited to symmetrical scaling on the range and can produce interesting pitch scaling by setting ranges on either side of the zero point to different offsets. Perhaps not that useful for traditional tuning but an interesting option you can experiment with.



Note that like Y on Continuum, X retains its value at finger lift position but because you normally turn notes on and off with W gate or Z envelope returning to zero, the persistent value of X has no effect when notes are not being output. When a new note is output the current position of X is then used.

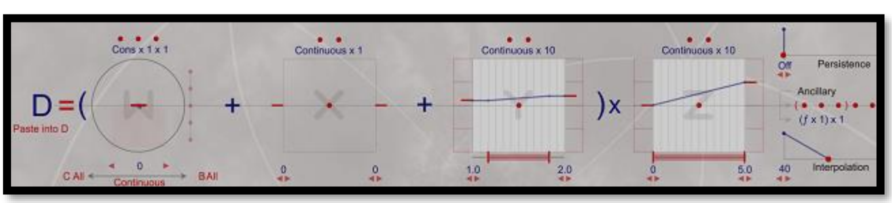
<u>Y CV Output</u>:

Here Y uses another 2-step function but in this case lower half of Y will output 0V and upper half of Y will output 1V. This approach is useful for creating switching applications. Because Y retains its value, when you lift your finger in the top half of Y, the CV output for Y will continue to be 1V until you press in the bottom half of Y creating a touch sensitive voltage output switch.



<u>Z CV Output</u>:

Here Z will output 0 to 5V as you press to max Z while Y is at min (0). Y is shelved a bit here so you can easily create the Z only value. However, as you raise Y from bottom to top, Z is multiplied by an increasing value from 1.0 to 2.0. So at Max Z pressure and at Top of Y (which is also shelved for ease of manipulation) the output will be 10V (or very close to it) and some fraction between 0 and 10V when Y is between 1.0 and 2.0. This is another example of a mixed formula, here using Y and Z though only Z value will be output as the formula is on the CVC Z output. Also note that Y multiplication was used instead of addition to avoid Y retention issues when lifting your finger as noted above. Since Z will always return to zero on lifting your finger the value of Z will always go to 0V in this case, no matter what the value of (W+X+Y) in this formula might have been.



8.7. CVC Programming Use Case 3a – EaganMatrix Module as LFO Generator

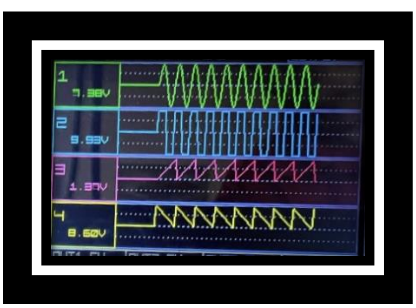
While the EaganMatrix Module requires its Y and Z CV inputs to be -5V..5V signals, you can program the CVC Y and Z outputs to support the full -10..+10V range of each CVC port. You do not have to map how the module uses its W, X, Y and Z inputs to play internal presets to the same CVC output functions if you choose to use the module in a different manner.

Let's say we want to control the EaganMatrix modules though normal monophonic CV inputs but we want the CVC to output a variety of LFO signals when the module receives a Gate >=1V at its W gate input (no other CV inputs are required). In this case we are using the module as a CV-based LFO generator and are not playing it.

- Set Polyphony = 1 (Outputs on CVC channel 1 – Ports 1, 2, 3 & 4)
- While Module is Gated:
    - For W, output a bipolar continuous Shape Generator 1 (SG1) based LFO Sine Wave from -10V to +10V at a frequency controlled by Macro Controller i and width controlled by Macro Controller iii. Change the frequency in continuous (small) increments.
    - For X, output a bipolar continuous Shape Generator 1 (SG1) based LFO Square Wave from -10V to +10V at a frequency controlled by Macro Controller i and width controlled by Macro Controller iii. Change the frequency in continuous (small) increments.
    - For Y, output a unipolar continuous Shape Generator 2 (SG2) based Ramp Up (sawtooth-like) Wave from 0V to +10V at a frequency controlled by Macro Controller ii and width controlled by Macro Controller iv. Change the frequency in increment of .5 seconds.
    - For Z, output a unipolar continuous Shape Generator 2 (SG2) based Ramp Down (sawtooth-like) Wave from 0V to +10V at a frequency controlled by Macro Controller ii and width controlled by Macro Controller iv. Change the frequency in increment of .5 seconds.

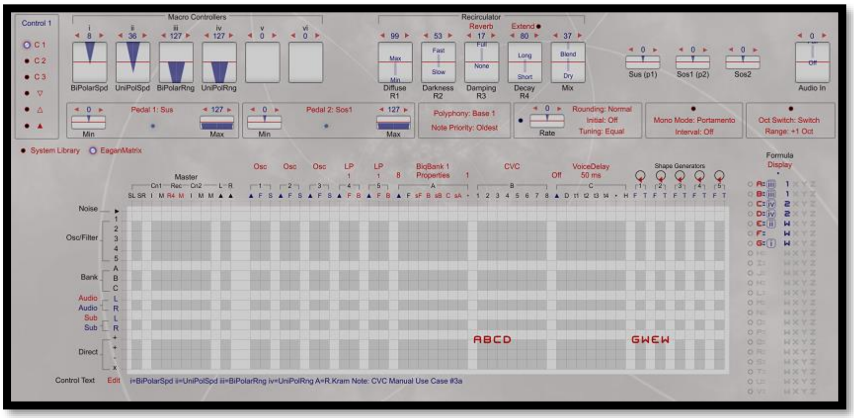
  Note: Here there are four LFOs, two bipolar and two unipolar, working in pairs to optimize use of only 4 Macro Controllers. Each pair has the same frequency and width controls as we want to control this with the 4 Macro Controller knobs available on the module's front panel.

Here is the scope output given one set of Marco Controller speed settings at full width setting:

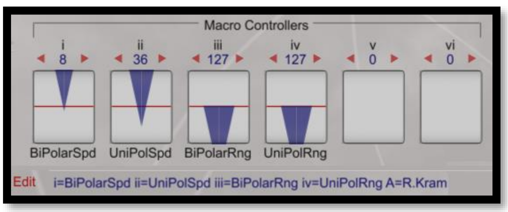


Note: The clocking output of the SG may not be as accurate as a Eurorack LFO or timing module intended for high speed clocking. This is a programming example to show flexibility of the CVC interface. If you need exact LFO timing, please use a module intended for that purpose.

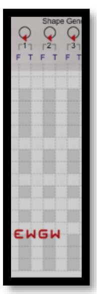
The Basic EaganMatrix program to create the LFOs is simple from the top level:



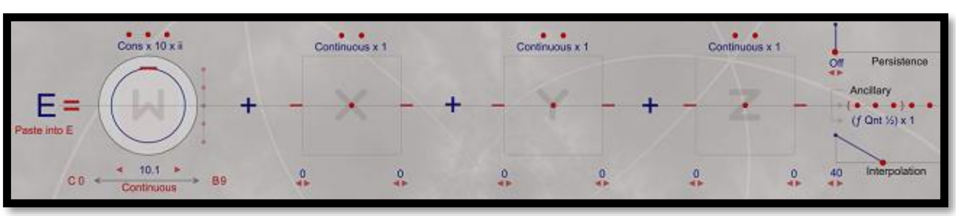
Four Macro Controllers are defined to control Speed and Width/Range of the pairs of unipolar and bipolar LFOs. The following setting maps to the scope trace above.



Two Shape generators are defined to control the speed of the ramping and the square pulsing. The frequencies of each are set using formulas E (for bipolar LFOs) and G (for the unipolar LFOs).

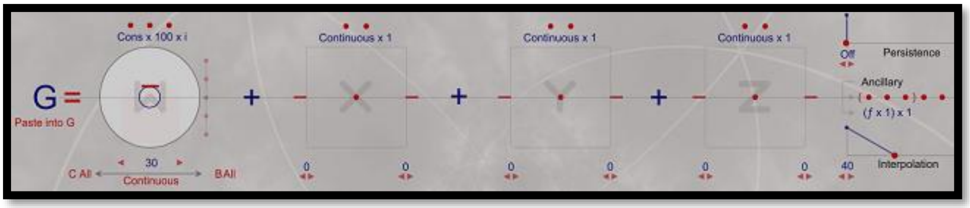


SG2 Frequency Formula E: Sets Speed of the unipolar ramping from 0 Hz to 10 Hz in increments of 0.5 Hz (seconds). Sets W with max value 10.1 with Ancillary Qnt (Quantization) by 0.5. Values will range from 0..10. Notice that the range of W is set slightly above 10 (10.1) to make sure the quantization reaches 10 (if set at 10 it will max out at 9.5). Control of this is associated with Macro Controller ii.

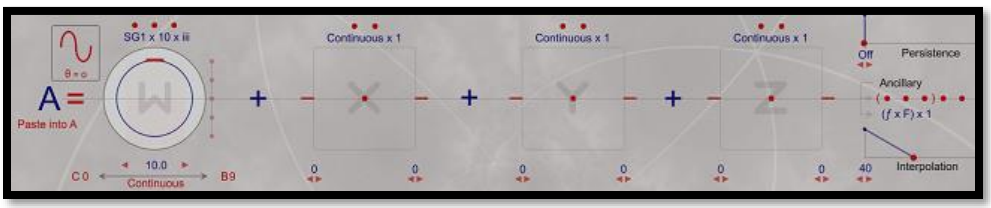


SG Frequency Note: Shape Generators should not be set with frequencies > 100Hz.

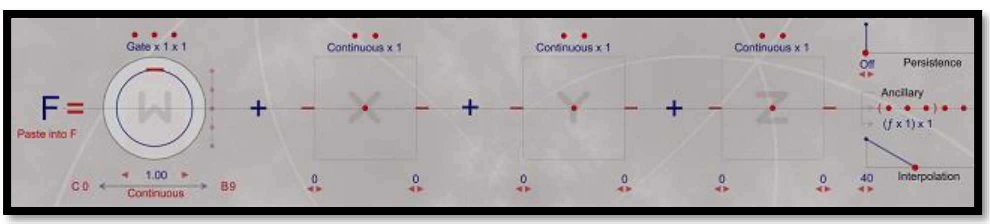
SG1 Frequency Formula G: Sets Speed of the bipolar sine and square LFO outputs. Here the range is set 0..30 Hz (seconds) and no quantization is used so increments of frequency change will be in units of 1/128 as the Midi range of the Macro Controller is 0..127. Thus, both LFOs are not super-slow or super-fast (you can change the W constant value to create slower and faster ranges). Control of this is associated with Macro Controller i.



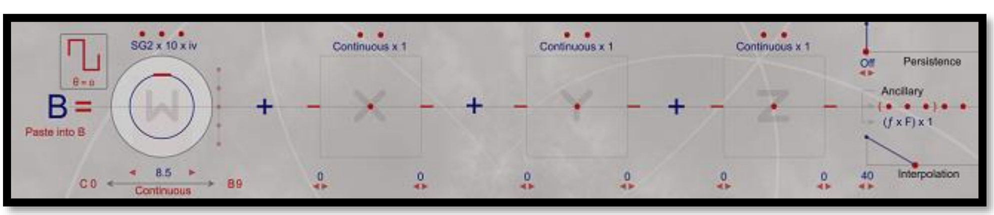
W Programming (formula A): Sets a continuous SG1 based shape generator set to bipolar sine shape with Max W of 10.0 (Volts) controlled in range/width by Macro Controller iii. This will output the sine LFO in the range -10V to +10V as it is interpreted as a bipolar output at the frequency set by Macro Controller i (actual output may be 9.95V, etc.).



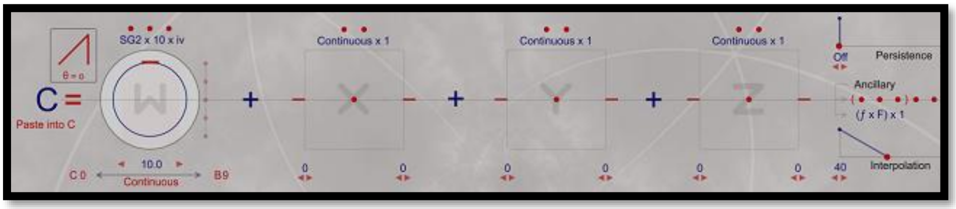
Note an Ancillary multiplication by formula F is used. This applies a gate control to Formula A so that the LFO wave is only output when the module is gated. All the wave output formulas use this Ancillary gating formula as we only want the LFO to trigger when the W input to the EaganMatrix Module sees a >=1V signal.



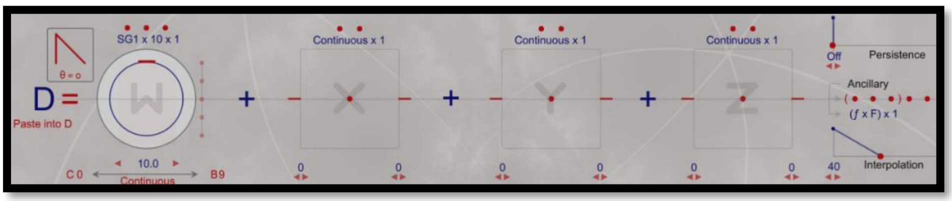
X Programming (Formula B): Identical to formula A except continuous SG1 is set to the bipolar square shape. This will output our square LFO using the same controls as the Sine LFO. Note this has nothing to do with X on the fingerboard or an Osmose key frequency. X is here independently programmed to output the LFO voltage. CVC outputs do not have to map to traditional W, X, Y, Z functions.



Y Programming (Formula C): Formula C is similar to Formulas A and B only it is set to continuous SG2 using a unipolar ramp up shape. Setting W to 10 will create a ramp of 0 to 10V in this case with range/width controlled by Macro Controller iv and Frequency controlled by Macro Controller ii. Formula F gating control is applied.



Z Programming (Formula D): Formula D is identical to formula C except it sets SG2 to a unipolar ramp down shape. Range/Width and frequency controls are the same as for Y LFO output.



The preset described above, including the second set of controls for Pulse and Ramp width, is available at the EaganMatrix PatchStorage.com repository here: CVC Use Case 3a | Patchstorage

8.8. CVC Programming Use Case 3b – Simple Envelope Generator with CVC

A similar CVC programming preset approach to the above LFO output is using single cycle shape generators instead of continuous and modifying things to output four different envelope shapes that can be used as VCA inputs to things.

As in the last example all we need is to trigger the module though a gate and the programmed envelope will be applied to the desired CVC output ports. This could be a gate from a CV keyboard as in the figure below, or you could use a MIDI note on as a gate if you are connected to the Midi port.

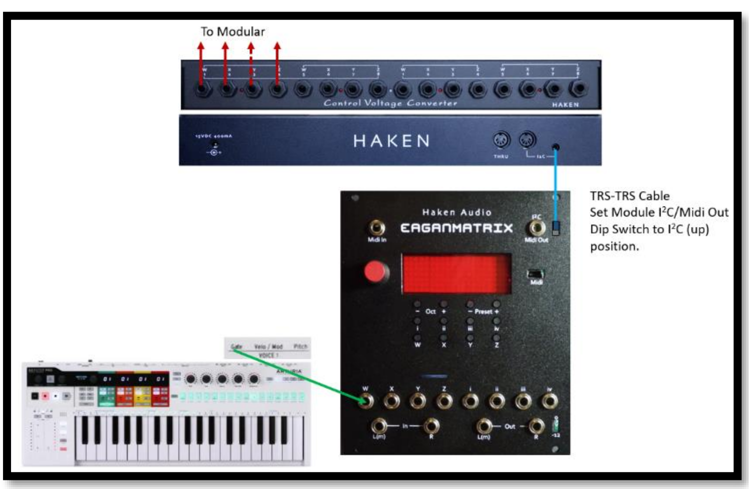


Figure 17- Using EM Modules and CVC as Simple Gate Driven Envelope Generator

True ADSR envelopes are difficult to program in the EaganMatrix so we will create a preset that outputs CV for an ASR envelope - attack (single cycle shape generator) at some frequency set by Macro Controller i, sustain (the time you continue to apply the gate while finger is pressed) and release (time of release phase using Blend of persistence after finger is released). This is a common method of creating envelopes for internal presets. Here we apply it to CV output. For simplicity this preset will only create the envelope on the Z output but you could create different envelope shapes on each of the outputs if you desired.

This preset does the following:

- Set Polyphony 1 so the outputs will occur on CVC ports 1-4 as in the previous example.
- Set Z to ramp up attack. Ramp it at Speed controller by Macro Controller i. Ramp it to max level controlled by Macro Controller ii.
- Hold note at the ramp point through the gate.
- On Release (removal of gate) create an envelope release though blend persistence with release time set by Macro Controller iii.

Note: An envelope decay phase could have been added by incorporating another Shape Generator triggering when the first reaches it maximum level but for simplicity we will leave that as an exercise for the reader if they desire to attempt it. There are a few system ADSR-centric presets you can consult.

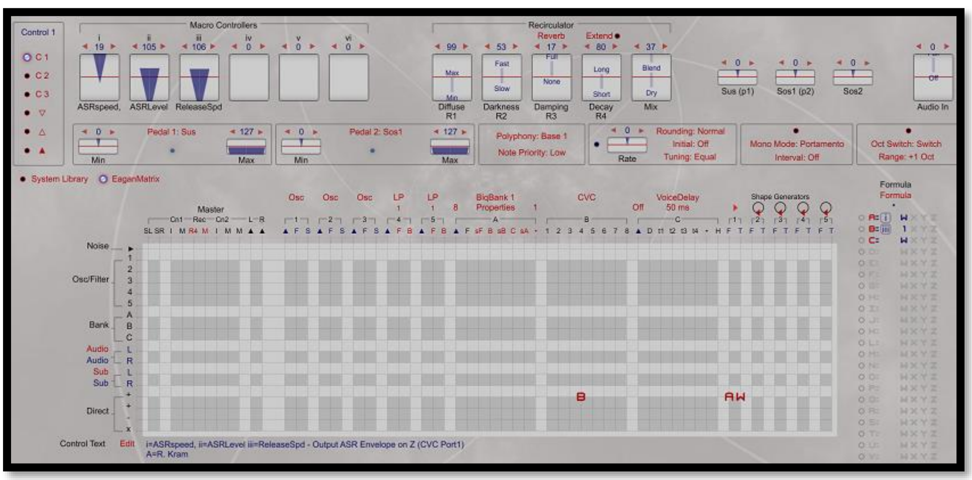


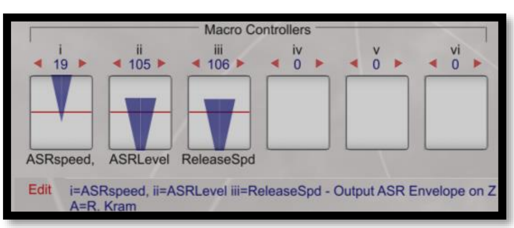
Figure 18- Simple ASR CV Envelope Output on Z

Let's start by defining three Macro Controllers:

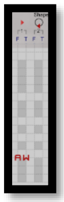
i – Sets envelope ramp up speed

ii – Sets envelope max range assuming you hold gate long enough to reach it given the current ramp speed.

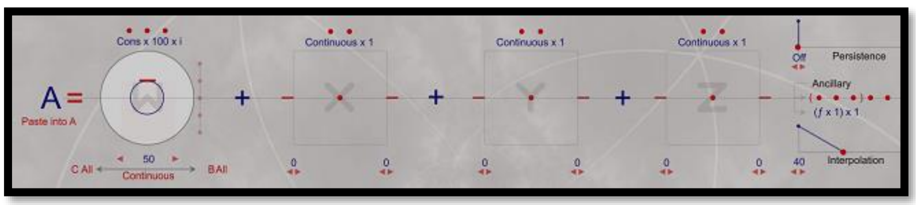
iii – Sets release time (Persistence output)



First, define Shape Generator (SG1 here) frequency and set to single cycle instead of continuous as was done in the last example. We want a single ASR voltage output per press.

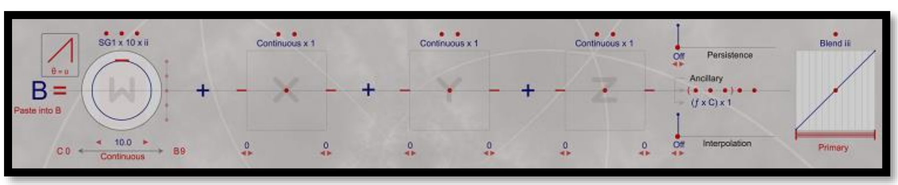


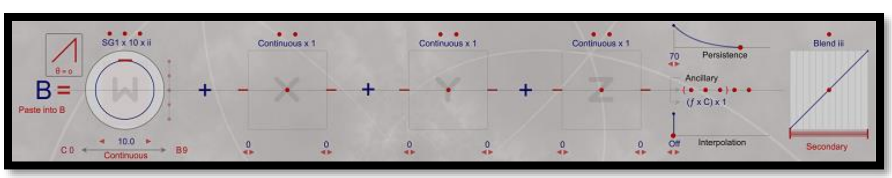
Create a formula to control the frequency of SG1 – here sets from 0 to 50Hz (of course you could set this to a slower or faster max speed if desired). Speed is set by Macro Controller i.



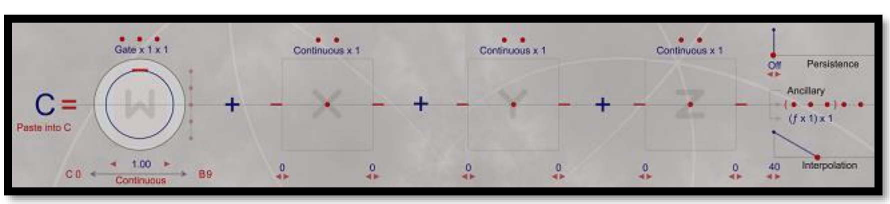
Then we create the SG Attack ramp up formula with maximum value of 10V, range controlled by Macro Controller ii and speed of attack again controlled by Macro Controller i. As long as Gate is held the CV will be output at its set maximum range.

When the gate is removed (finger lifted if that was gating) the Release component kicks in using the blend of Persistence of Macro Controller iii which controls the rate of release (amount of persistence applied). Here the primary is set to zero (immediate release) and the Secondary reaches a maximum persistence of 70 (a relatively long release but you can set this lower or higher to your liking). As you bring up Macro Controller iii the release time gets slower.

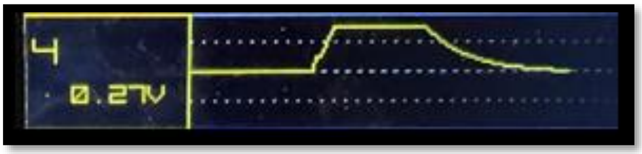


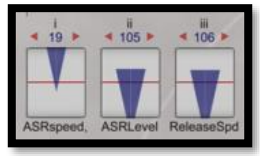


As in the previous example we create an Ancillary Gate multiplier formula (formula C) to make sure the W Gate drives the envelope only on pressing.



Here's an example ASR CV output based on the settings above. Remember the ASR envelop is only being output on Z, our CV port #4. Change Attack frequency for a faster or slower attack. Change Persistence





Here are three successive triggers of the ASR envelope, each time holding sustain a bit longer, so your finger is a direct component of the envelope:

8.9. CVC Programming Use Case 3c – EaganMatrix Module as Midi-CV Converter with CVC

The EaganMatrix module will output the midi input translation to the CVC port if it is set for CVC connection. In this case the module can function as a MPE Midi to CV converter for an external Midi source. For example, let's play the module from a Linnstrument and send the Linnstrument's CV output (which it can't support without an MPE-CV converter) to other Eurorack CV inputs so that we can simultaneously play the EaganMatrix module and up to 4 Eurorack Oscillators in parallel (though we will use a single quad oscillator in this example).

Here is one possible setup using a 4-voice (Synthesis Technology) Wavetable Oscillator with a 4-channel (L1) Quad VCA/VCA and 4 channel XAOC Praga Mixer. Of course, you could have connected to four separate Oscillators as well, etc. The Pressure envelope of the Linnstrument will drive all the CV connected oscillators through the VCA.

The Linnstrument is set up to output 4 MPE (Voice per Channel) voices to the EaganMatrix Module which when connected to the CVC will be converted (if played with 4-voice polyphony) to the 4 CVC channels of W, X, Y and Z CV outputs available. The Quad Oscillator sends its audio output to the Quad VCA which uses Z for creating the Envelopes generated by the Linnstrument. W gate is not required. This is similar to how you would send your Continuum envelope to your modular gear if you are using that to control your setup. Of course you can also use the Gate output from the Linnstrument if you choose which maps to Note on/Note Off Midi messages.

Note: This setup would work identically for Linnstrument connected to Continuum, ContinuuMini or Osmose as well.

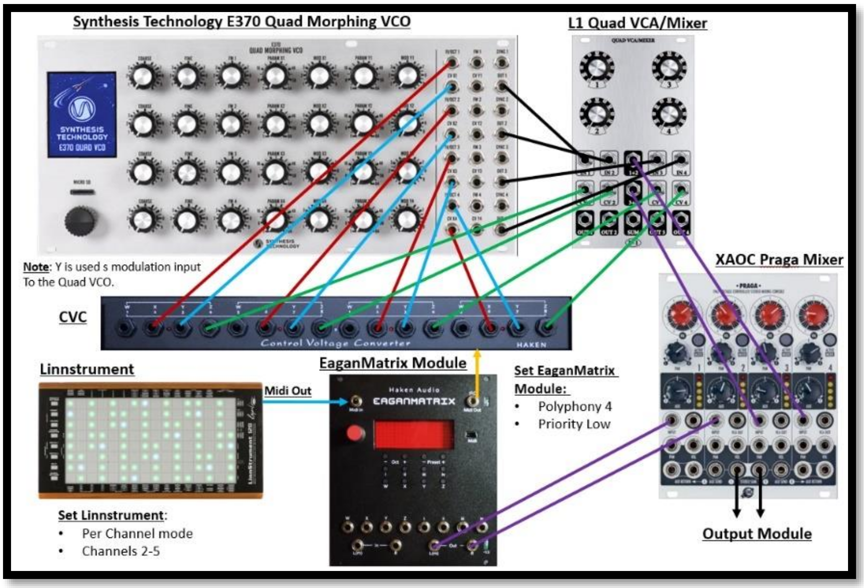


Figure 19- Using Linnstrument to Play Eurorack Modules Through 4 Voice CVC Connection

System Setup

This is just one of an infinite set of possibilities for Eurorack control. In this case you do not need to write a CVC program, you simply do the following:

- Option: Load an Empty preset on the EaganMatrix Module that is connected to the CVC as detailed above. This assumes you do not want to play the EaganMatrix module with the Oscillator. If you wish to play the module in parallel with CVC output, load the desired preset and set it for the same configuration below.
- Make sure no CVC bank is defined in the EaganMatrix as this is not programmed as in the last few use cases.
- Set your Polyphony to 4 on the EaganMatrix Module.
- Set Priority Lowest so you can predict which channels play with what fingers as you play polyphonically on your Linnstrument
- Set the CVC Setup to one of the 7 CV output configurations to match the needs of your target oscillators and or other CV targets.
- Connect your Linnstrument to the module's Midi TRS input port with a DIN-to-TRS adapter
- Set your Linnstrument to play Per Channel mode as if you were going to play the EaganMatrix module through Midi (see Continuum Manual for setup). Set to play four channels -2-5. Set CC74 output if you want to use it. Set to Channel pressure (see Continuum Manual or Continuum Interfacing Cookbook for Linnstrument setup for playing the EaganMatrix Module (will be same as for playing Continuum or ContinuuMini).
- Now when you play your Linnstrument your MPE Midi output will play the EaganMatrix module if you have loaded a preset and the MPE data will be converted to CVC output and sent to the four CVC channels of W, X, Y and Z per the Lowest priority (you can change that priority if you have a desire to).
- Connect the CVC outputs to the desired modules to play your modular polyphonically over MPE from the Linnstrument.

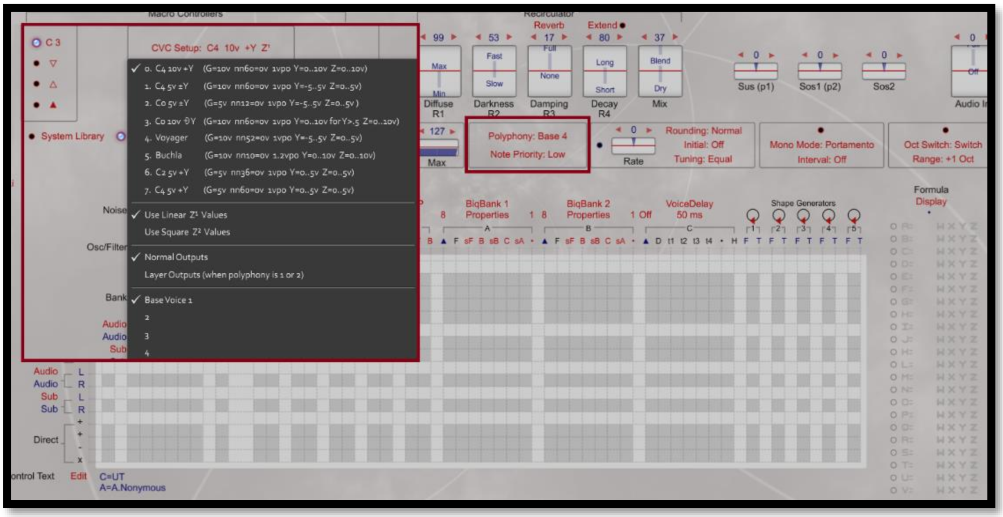
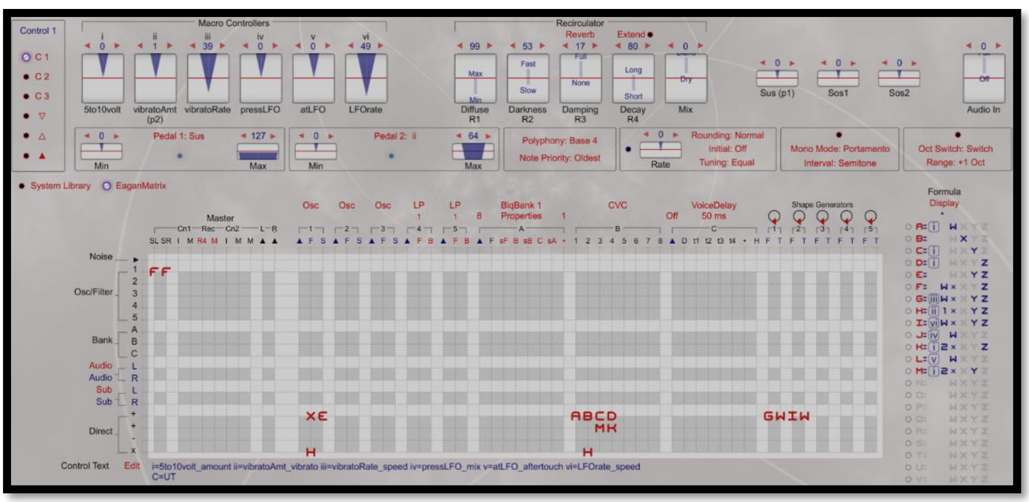


Figure 20 - MPE Midi to CV Conversion Example

8.10. CVC Programming Use Case 4 – Osmose ("cvc 4 voices" preset)

Now let's look at one of the Osmose System Presets that are meant for CVC use in the Utilities Category: "**cvc 4 voices**". This is more complicated than the EaganMatrix module examples above and as normal for Osmose presets uses all six Macro Controllers to adjust the CV output in various ways. It is a good illustration of how you can create sophisticated CV output from a relatively simple EaganMatrix program.



This preset does the following:

- Output Osmose audio as well as sends output to the CVC
- Set Polyphony = 4, priority = Oldest. So ports 1-4 (W, X< Y, Z) output will rotate using Oldest priority CV between those ports and ports 5-8, 9-12 and 13-16. Set this Lowest priority to achieve a more predictable CVC channel output based on key pressed and polyphony played.
- **W** Gate outputs a configurable voltage from 5 to 10V controlled by Macro Controller I (5to10volt).

- **X** CVC output is set to output 1V/Oct scaled to Octave mode with Zero Point set at C3 (appropriate voltage will be output to support that but of course if sent to a modular Oscillator you would still normally have to adjust the Frequency to match the scaled output). It’s assumed you would do that by playing a note on the Osmose and then with a tuner adjust your Oscillator to match the desired Zero Point frequency.
  Y+Z scaled Vibrato is also added to the X output based on Macro Controller ii (vibratoAmt) and iii (vibratoRate).

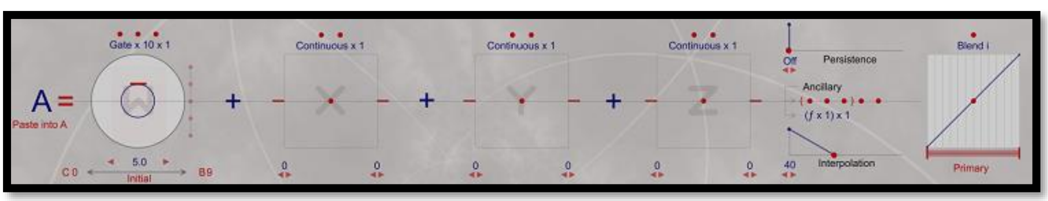
- **Y** CVC output is programmed to output a Linear Y voltage transfer function from 0 to an adjustable range of 5 to 10 V through Macro Controller i (5to10volt).
  Continuous Triangle-shape LFO modulation is added to CVC Y output (controlled by Osmose Aftertouch) with rate controlled by vi (LFOrate) and modulation width controlled by v (atLFO). Note **Midi Clock** also is used to adjust the LFO frequency.

- **Z** CVC output is programmed to output a Squared Z voltage transfer function from 0 to an adjustable range of 5 to 10 V through Macro Controller i (5to10volt).
  Continuous Triangle-shape LFO modulation is added to CVC Z output (controlled by Osmose Z squared pressure) with rate controlled by vi (LFOrate) and modulation width controlled by iv (pressLFO). This is then almost identical to design with the Y LFO output only Controlled by Osmose Z pressure instead of Osmose Y aftertouch.

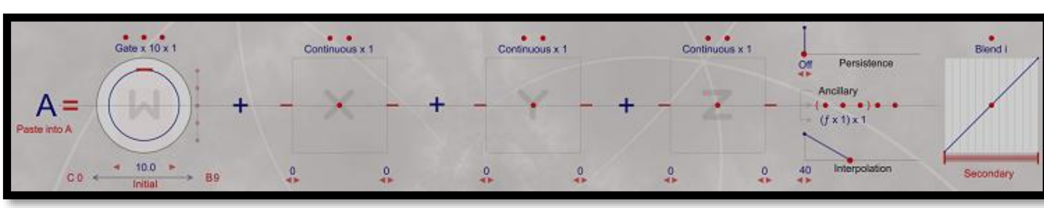
- Finally, An OSC is output to SL/SR so the preset makes a sound in addition to outputting CVC data. This is useful for testing your CV output as this uses the same vibrato control used by the X CVC output.

Programming W Port Gate CV Output

W gate is output using formula A incorporating a blend where the Primary is set to 5V to blend on i (5to10volt):

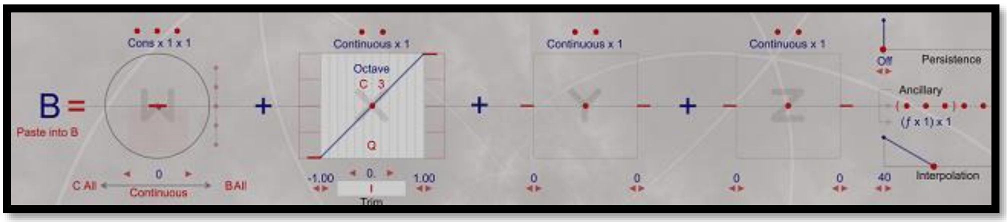


And the Secondary is set to 10V, so Macro Controller i will set the gate on value to an output voltage between 5 and 10V when Osmose key gate is applied (basically Note On). Note that on Osmose the sensitivity of this can be set for an almost imperceptible touch to a much firmer touch.



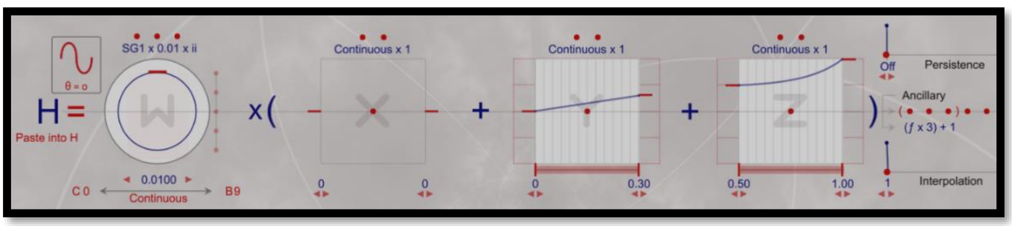
Programming Base X Port Pitch CV Output

Here the base X output voltage defined in formula B will map to the Midi Note played on the Osmose with a Zero Point of C3. This uses the standard X formula in Octave mode as the output is desired to be scaled according to Midi note, not a frequency in kHz. This octave scaling will cause 0V to be output when you play a C3 on the Osmose. C4 = 1V, C5=2V, C2= -1V, etc. as 1V/Oct scaling will be used as the default. If this is sent into an Oscillator as 1V/Oct pitch, you will need to calibrate the oscillator with a tuner to set it to the desired output frequency to match the Osmose/CVC output.



Programming the Configurable Vibrato on X

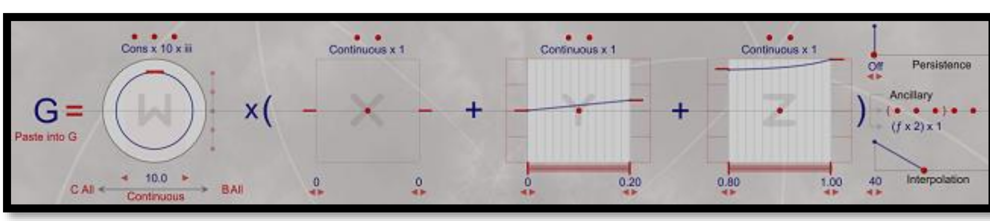
Formula H applies vibrato (SG1 Sine based frequency modulation on the X multiplication row) to the X pitch output which is scaled by Y+Z – a common Osmose scaling technique to apply a scaling through the entire key press.



The width of the modulation is controlled by Macro Controller ii (vibratoAmt). When this is set to minimum (zero) no modulation will be applied. Note the Ancillary formula (f*3)+1. When control ii is at zero the multiplication zeros out the formula but 1 is added so at min setting of i, H = 1 which is a unity frequency multiplier which in turn outputs whatever the value of formula B is.

Turn up control ii (assuming control ii sets a SG1 frequency > 0) and the maximum width for the sine frequency modulation will be 0.01 * (max value of X+Y = 1.3 at full Aftertouch pressure) * 3 + 1 = -0.96 to 1.04. So the actual modulation applied is quite small and will sound like a vibrato which you can control the width of. Because the same vibrato is applied to the OSC 1 audio output you can listen to that to get a good idea of what your CV modulation will be when applied to your modular gear in some way.

The rate of this modulation is also controlled by formula G which controls the frequency of SG1 that is used to create the vibrato.

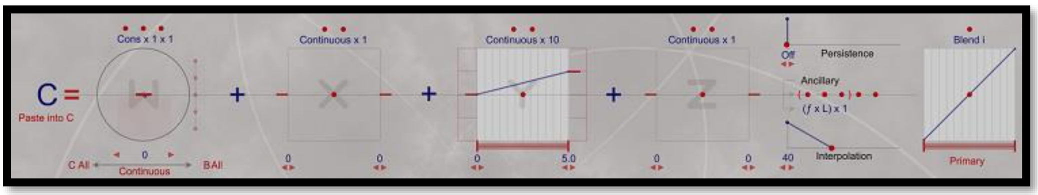


This uses Control iii (vibratoRate) to change the base frequency from 0 to 10 Hz, scaled by Y+Z and then multiplied by 2 in the Ancillary. When Control iii is minimum=0, frequency is zero and no modulation will be applied, even if Control ii (vibratoAmt) is set. The maximum frequency (value) of formula G when Y is at its maximum aftertouch pressure is then 24 Hz (10 * (0.2+1.0) * 2). This is a reasonable Shape Generator rate. Frequencies up to 80Hz or so are common. Rarely will an EaganMatrix SG frequency be set > 100 Hz.

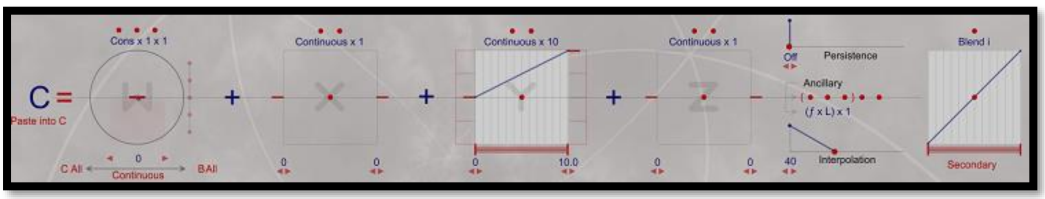
Programming Base Y Port CV Output

The base Y port CV output uses formula C which outputs Y as a linear function scaled from 0V to 10V controlled by Macro Controller i (5to10V) with a blend function in a manner similar to how the Gate was scaled in voltage. Thus Controller i will set the range of Y output from 0 to the max blend value applied (10V).

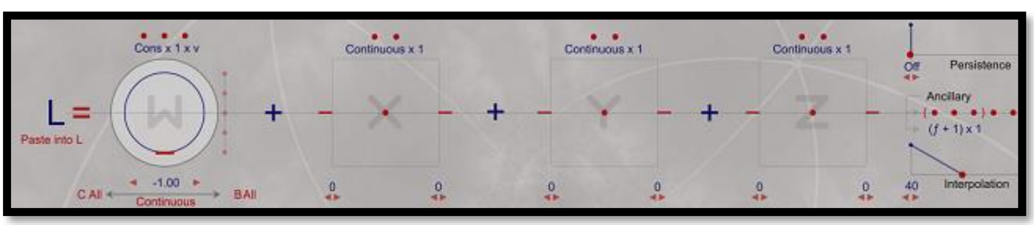
Primary Blend Formula – Y max = 5.0



Secondary Blend Formula – Y Max = 10.0



Note that Ancillary multiplier formula L is used in both blend formulas. This applies Macro Controller v (atLFO) that moves W in formula L from 0 to -1. Note the Ancillary in formula L adds 1. So when you move Controller v from min to max, the value of L goes from 1 (0+1) to 0 (-1+1). Since formula L scales formula C, L effectively decreases the value of C from its current value down to zero as you bring up Controller v.

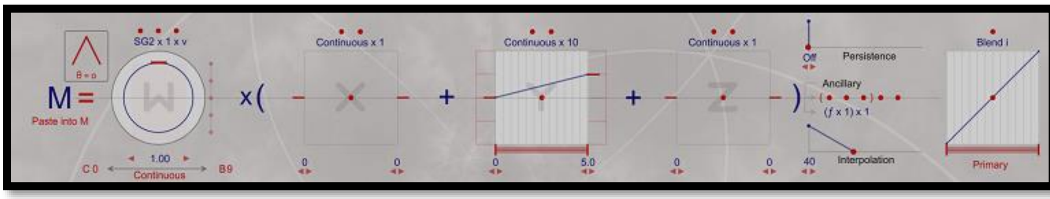


Note: On Osmose you can set internal Z and Y transfer function curves from 0 to 100 where 0 = maximum squared shape, 64 = linear shape and 100 = maximum square root shape. Since the output of the Y CV in this case is scaled as a linear function in formula C, you should set your Osmose to a similar Y shape so that the output is more predictable to Y pressure. For example, if you set your Osmose for Y=max transfer squared function shape, the output will be squared even though it is set linear in formula C. This would not be an issue on Continuum devices as they will use the Y shape in formula C directly. It's always best to set your Osmose Y and Z curves to Linear if you want to predictably match the CV output functions.

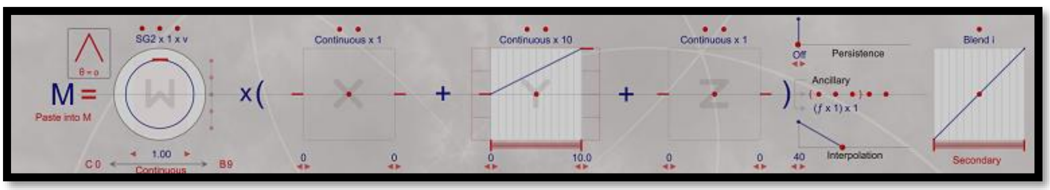
Adding Y LFO modulation

Formula M applies a SG-based triangle shape modulation to Y, with the modulation width scaling factor (W) controlled by Macro Controller v (atLFO). It uses blend using Controller i (5to10volt) to control the Y imposed range of modulation from 0 to 5V at Max Y with Control i at min and ranging to 10V when the secondary blend max is reached. So the total range of unipolar triable modulation width will be 0V to 10V (Scaled by W value). Only Osmose Aftertouch (Y) value will affect the voltage. When Aftertouch is not reached Y on Osmose will be zero and thus no modulation can occur until Aftertouch kicks in in this case,

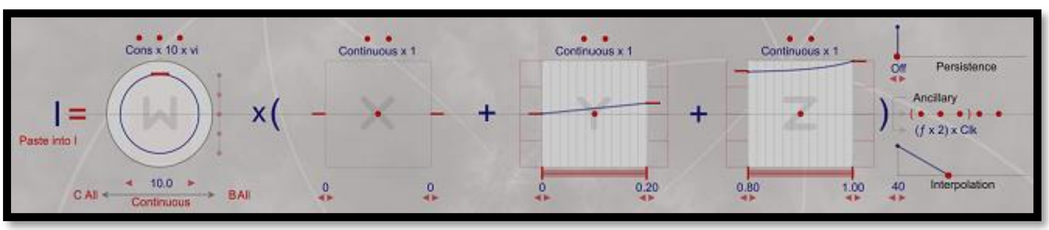
Primary Blend starting at max Y=5.0V when I (5to10volt is min). Scaled by W value controlled by v (atLFO).



Secondary Blend ranging to Max Y=10V when i (5to10volt is max). Scaled by W value controlled by v (atLFO).



Finally, the frequency of SG2 that controls the Triangle LFO uses formula I. The resultant LFO frequency is the value of W ranging from 0 to 10 controlled by Macro Controller vi (LFO Rate), scaled by Y+Z which take a max value of 1.2 (scaled identically as in formula G that controls rate of SG1). This uses an ancillary multiplying by 2 as in formula G, but note that is multiplied by incoming Midi Clock. So the exact frequency of the LFO will be dependent on Midi clock.

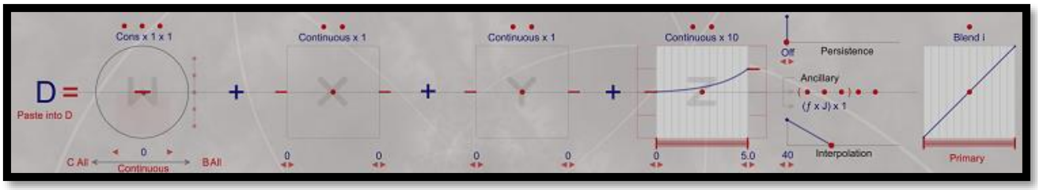


Programming Base Z Port CV Output

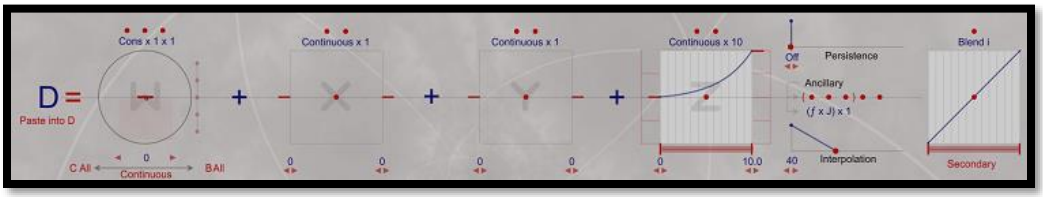
Z port CV output is programmed similar to that of Y, only Osmose Z pressure controls values instead of Y Aftertouch.

The base pressure value is set by formula D which uses Osmose squared pressure ranging from 0V to 10V – with a Blend to range between the two as in formula Y.

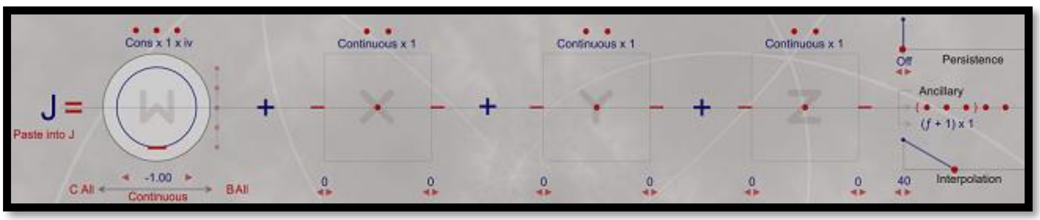
<u>Primary Blend starting at max Z=5V</u>.



Secondary Blend ending at max Z=5=10V



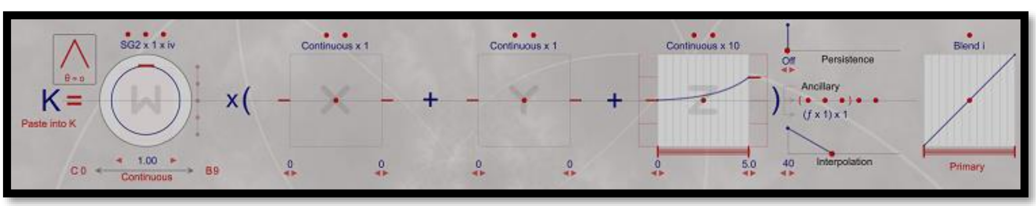
Formula D is scaled by formula J using the -1.0 offset similar to that used in formula L, only here controlled by Macro Controller iv (pressLFO). This reduced the value of formula D from its current value to zero as Control iv is brought to maximum.



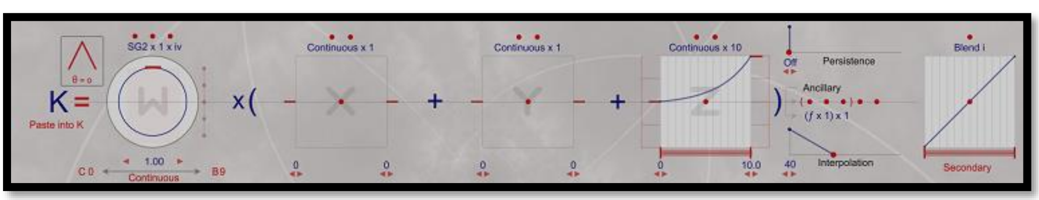
Adding Z LFO modulation

Formula K adds Z LFO offset modulation to formula D in a similar manner that formula M functioned with formula C for Y offset modulation.

Primary Sets range of modulation width control to Max Z = 5V when Control I (5to10Volt) is at maximum



Primary Sets range of modulation width control to Max Z = 10V when Control I (5to10Volt) is at minimum



This Z LFO modulation also uses SG2 so its frequency is set identically with formula I (also dependent on Midi Clock).

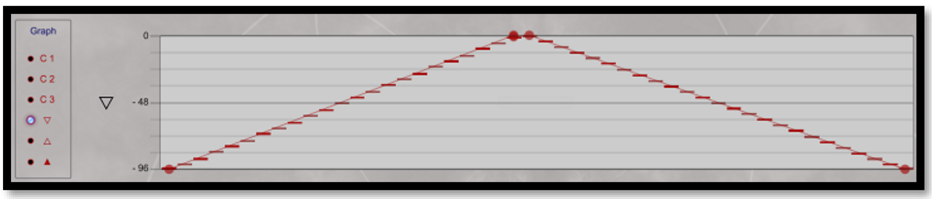
8.11. CVC Programming Use Case 5 – Using Main & Offset Graphs

The relatively recent introduction of Offset graphs for general programming (in addition to their original for BiqGraphs) opens up a world of arbitrary function generation of CV outputs either tied to X, Y or Z movement or tied to Shape Generator based continuous (voltage sequencing) or single cycle (envelope generation). A few simple examples will be presented here to over basic concepts of operation.

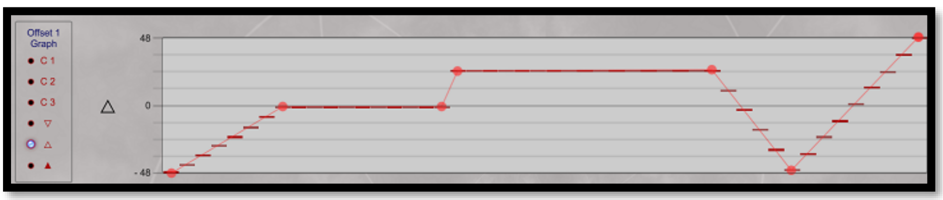
There are three graphs available in the Control panel – Main, Offset 1 and Offset2 graphs. See the EaganMatrix User Guide for general programming information and how to create the graphs.

Using the Graph Configuration tools, lets create:

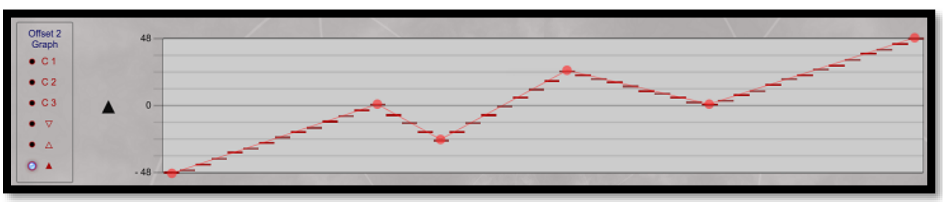
a) A Main graph as a simple triangle function:



b) An Offset 1 graph as an arbitrary function with different internal components:

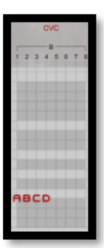


c) An Offset 2 graph as a zig-zag shape:

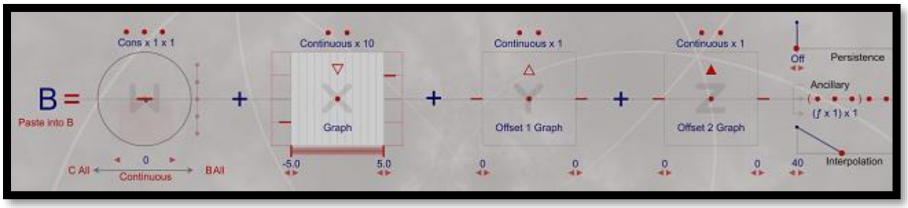


We'll use the full range (height and length) of the graph to maximize range of voltage output. For this use case we are not concerned with the graph's value because the scaling of the voltage output will be done in the X, Y and Z formulas that get assigned to each of these graphs. The voltage output will follow the shape of the graph using a range set in the formula driving each graph. These graphs are built with linear components but you can also add curved shapes as well with the building tool though the output is still discrete steps.

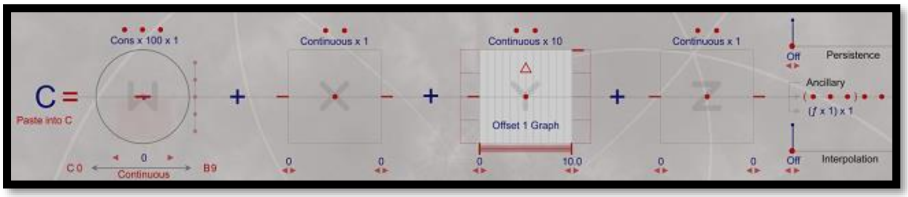
Our CVC program uses ports 1-4 with formulas A (Gate), B(X), C(Y) and D(Z) set Polyphony 1.



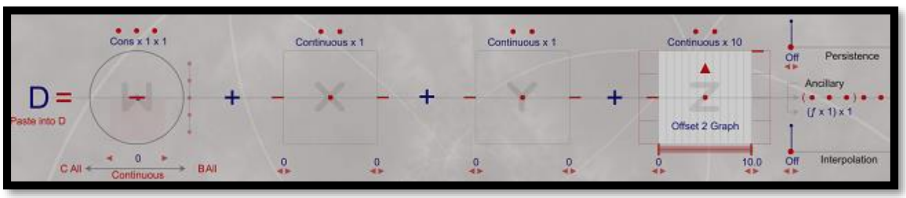
For formula B we'll assign X to the Main Graph with a scaling of -5V to +5V.



For formula C we'll assign Y to the Offset 1 Graph with a scaling of 0V to +10.



For formula D we'll assign Z to the Offset 2 Graph with a scaling of 0V to +10V.



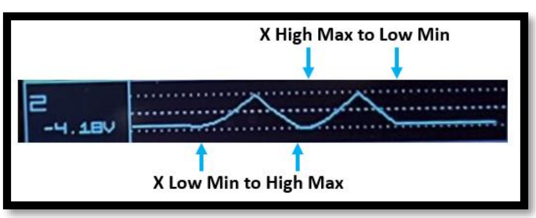
Scaling Note: When using Offset graphs, it's a good idea to use a 0V to max 10V scaling which will map to the graph as expected. Also Interpolation is turned off as linear interpolation should be applied see smoothing note below.

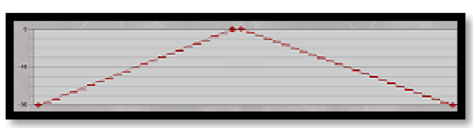
<u>X Operation</u>:

As you run a portamento from bottom to top of fingerboard, or from top to bottom because the shape is symmetrical, you will get voltage function output as a triangle shape from -5V to +5V in this case. The voltage rate of change will match the speed of your portamento.

Note: Since X remains at its last value when you lift your finger, if you stop in the middle of the function the voltage will remain at that point.

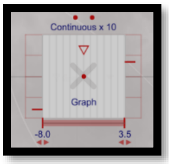
Here a scope output running from bottom to top and then top to bottom for the X Graph function:





Note: Starting in the middle of the fingerboard and creating a portamento will create a different output and scaling. Experiment if you like. The advantage of using the X Graph is you have so much room to travel from low pitch to high and back that you can get more control of the output. Of course, in this case you are not using X for pitch in the traditional sense. Simply increase the range of X to impose a larger voltage range. And the scaling does not have to be symmetrical.

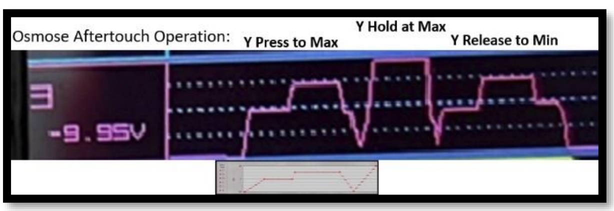
This function is not really useful on the Osmose that cannot produce true portamento. You could try applying a pressure weighted portamento while connected to the CVC, but that is very difficult to maintain an even rate to produce the desired voltage shape. You could range the output to a larger scaling if you like, and the range does not have to be symmetrical. For example, the X range for the graph's output voltage could be from -8V to +3.5 volts:

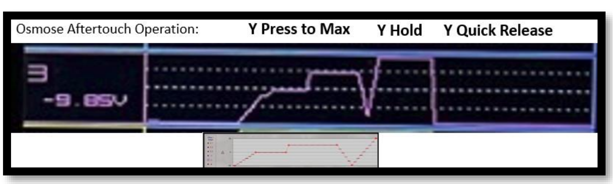


Y Operation:

As you move from bottom of the Continuum to top, Y will output its graph as a voltage pattern in a similar manner. However, Y does not have the same resolution as X or Z as on continuum it is output as an average value based on position. It is not recommended that you use Y as a voltage graphing function on Continuum unless you are ok with the distortion that will result because of its accuracy and ContinuuMini has such a small Y range to work with it is also not recommended for graphing use.

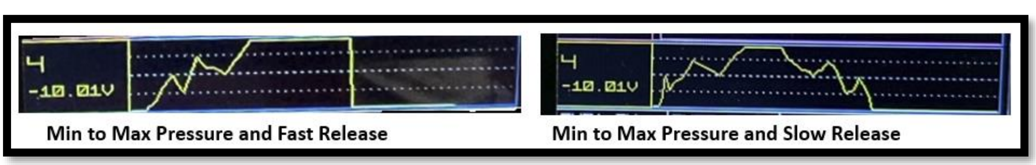
However, Y on Osmose is a different story. That is more controllable and returns to zero so it is a better option. One you reach the bottom of the Aftertouch area, on return to zero (key release) the shape will be output in reverse if finger is release slow enough. If the finger is quickly released the voltage will almost immediately drop to zero. This takes a bit of practice to get the shape to output as desired but on Osmose it is rather predictable and might be very useful for creating interesting Aftertouch-based voltage functions/effects.



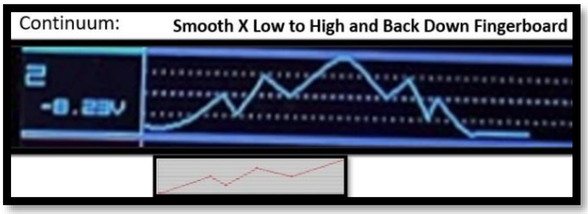


Z Operation:

As you press from min to max the voltage graph function will be applied maxing out as you continue pressing at maximum pressure. As in the Y example with Osmose, a quick release will cause the voltage to almost instantly return to zero. If you release slowly, the pattern will repeat in reverse over the voltage range define in Z. However, as you can see, it's very difficult to reliably recreate exactly the given voltage graph with pressure on Continuum.



Compare this to setting X with the same function (now scaled 0 to 10V) where it's easy to smoothly run your finger the fingerboard across and back at a relative constant speed:



Osmose also has sensitive Z pressure and perhaps a bit less easy to control than Continuum pressure which is a tad more sensitive and controllable but Osmose get very good graph response output for Z as well.

Osmose Sensitivity Settings: On Osmose it's a good idea to set your Z and Y (Aftertouch) pressure response as **Linear** in the menu system to get the most accurate voltage graph translation.

CVC Voltage Smoothing: Using this technique where internally control maps from 0..1 linear interpolation should be used to smooth out values. This will occur if you are using a graph with integer control (not discussed here – see EaganMatrix manual). If you find you are getting voltage stepping on output you can try turning up the interpolation control.

Summary: When played fast enough this technique allows you to create arbitrary Z-controlled envelope shapes. As you can see, it's difficult to create the exact shape without applying the other technique of outputting the graph using a shape generator with fixed speed, but the technique as applied to X, Y and Z can create very interesting shaped-voltage outputs.

The preset that accompanies this graphing discussion is available on the EaganMatrix Patchstorage.com site here: CVC Use Case 5 (CVC Graphing) | Patchstorage

8.12. CVC Programming Use Case 6 – Other Graphing Use cases

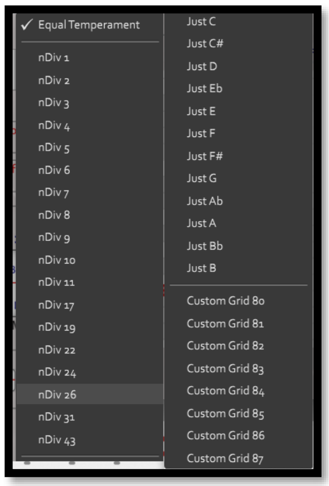
See the EaganMatrix User Guide for more information here and videos on the graphing functions on the YouTube EaganMatrix Programming Channel.

The graphing functions can be applied to shape Generator control to allow for:

- Creation of Note sequences using Continuous Shape Generators that will be translated to 1V/Oct voltage if sent out the CVC
- Creation of Arbitrary envelope shapes using single cycle Shape Generators primarily applied to Z
- Creation of continuous SG-based arbitrary functions applies to Y and Z.

8.13. Alternative Tuning Output through the CVC

The programming examples presented thus far assume equal temperament 1V/Octave output, however you can use the Pitch Table Editor to create alternative tunings and scales whose X scaling will be output to the CVC if X is associated with the new custom Grid. In addition, Just and nDiv options are available. As noted earlier, Rounding (on Continuum-based devices) can also be set and that will affect the X scaling and associated pitch bend that will be output from the X CVC port. See the Pitch Table editor and Continuum User Guide for more information. Note that some of these options may not apply to the Osmose when setting in the editor.



9. CVC Troubleshooting

Here are some common troubleshooting issues:

9.1. Blue Light does not come on when powering.

- If the Blue Led on your CVC does not light up when turned on, please check the external power supply.
- if you have an older CVC, power the unit off, disconnect the power cord and check the fuse.

9.2. One or more Red Lights are stuck / remain constantly on

- If the red lights do not all go off when not playing the fingerboard or Osmose keyboard and no external Midi is applied, power off the CVC, wait a few seconds and power back on.
- If this does not work, power down the device connected to the CVC as well as the CVC and then power them up (order should not matter).

9.3. CVC does not output Voltage

- Make sure your CVC is properly powered and connected to your Continuum with the appropriate cable for your instrument detailed in section 2.
- Make sure you are using TS (mono cables) to connect the CVC outputs to your CV destinations.
- Make sure the Red Lights activate as expected when you press the fingerboard or Osmose keyboard.
- In the editor, make sure you see the CVC Connected message in the CVC Setup control when hovering your mouse over the control.
- If connecting to a Continuum or ContinuuMini, make sure it is properly calibrated (does it play notes as expected mapping to the fingerboard). Try listening to the Continuum headphone output to verify that it is working. If your CVC is connected to your Continuum Fingerboard it must be powered up, else your Continuum Fingerboard's headphone output will not function and your Continuum Fingerboard's LED will flash violet to indicate the problem. You must either power up your CVC or disconnect it from your Continuum Fingerboard.

9.4. Older CVC does not work when connected

- Make sure you are using a fully populated Midi cable if connecting to the Midi DIN port
- If using a TRS adapter, make sure it is built correctly according to the pinouts specified in this document (See section 3).

9.5. Unexpected CV output

- If the Red Light output sequence is not working as expected first check your MPE Priority set and confirm it matches the expected operation

- Make sure you did not set CVC Base in the editor to more than 1 as that will limit output on CVC ports.
- Make sure you are not connecting to both the TRS and Midi port at the same time.
- If your CVC gate indicators show that it is not using the outputs – if only some, but not all, of the CVC gate outputs ever light up – the Continuum may have misidentified your CVC. Turn off your CVC (but leave your Continuum on), play some notes on the Continuum while the CVC is off, then turn the CVC back on and play some more notes. If your CVC's address is correctly identified in the Continuum Editor, then the problem is something else; carefully check the configuration of your Continuum: Polyphony, Channel Assignment, Split, and Mono Interval affect which voices are used.

9.6. Intermittent notes

- If you have intermittent problems with the CVC (notes drop out once in a while), you may be experiencing i2c communications errors on your Midi cable. Try a different high-quality cable, and/or a shorter cable.

9.7. CVC does not work with my xCEE

- If you have an xCEE you must Connect your CVC to Data Port 4 on the Expander; it will not work if you connect your CVC directly to your continuum. You will see a flashing violet LED if your CVC is connected but turned off, or if your xCEE is not communicating with your Continuum Fingerboard.

9.8. Osmose does not connect to the CVC

- Make sure the Global Midi I/O setting for "din Mode" is set to "5/5 CVC".
- Make sure you are using a fully populated Midi cable to connect to the CVC from the Osmose Midi output port.

9.9. Haken Editor does not see the CVC in CVC Setup

- Make sure the editor can talk to the Continuum and that you do not have CVC routing turned off
- Make sure CVC is connected and operates as expected when editor is not running and troubleshoot there as needed.

9.10. Pitch Bend X values do not seem to output or seem incorrect

- Make sure rounding is not on for the preset or set of or to a very low option on Osmose

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9.11. Osmose Sequencer data is not output as expected to CVC

- Make sure normal CVC operation from keyboard works as expected (Din Mode is set to "5/5 CVC").
- Make sure you have set your preset to Polyphony = 1 in the editor

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9.12. I can't connect two supported instruments to the CVC

- That is correct. You can only connect to either the Midi DIN or the TRS port.
- If you have multiple instruments that support the CVC, only one can be connected at a time.

9.13. Contact Haken Audio for Further support

- If any of these problems persist, please contact Haken Audio for further guidance by raising a support ticket at hakenaudio.com/support.

10. CVC Specifications:

SPECIFICATIONS

- **Communication Rate**: 440 kbaud
- **Interface**: I^2C link to Continuum, ContinuuMini, EaganMatrix Module, or Osmose (DIN5 and 3.5 mm)
- **Control Voltage Output**: Sixteen 16-bit D/As, each -10 V to +10 V (actual output may be slightly lower or higher)
- **Power**: 10 W, 110 or 220 VAC (country plug adapters included)
- **Size**: 38.1 x 15.3 x 4.4 cm (15 x 6 x 1.73 inches)
- **Weight**: 90 g (3 oz)

11. References:

See the following references for more information on EaganMatrix, Continuum instruments and Osmose keyboard.

- Continuum User Guide. Haken Audio: Editor, Firmware, and User Guides — Haken Audio
- EaganMatrix User Guide: Editor, Firmware, and User Guides — Haken Audio
- Osmose Online Manual: Online Manual - Help Center (happyfox.com)
- Continuum Interfacing Cookbook: Editor, Firmware, and User Guides — Haken Audio
- EaganMatrix Programming Cookbook: Editor, Firmware, and User Guides — Haken Audio
- Pitch Table Editor: Editor, Firmware, and User Guides — Haken Audio